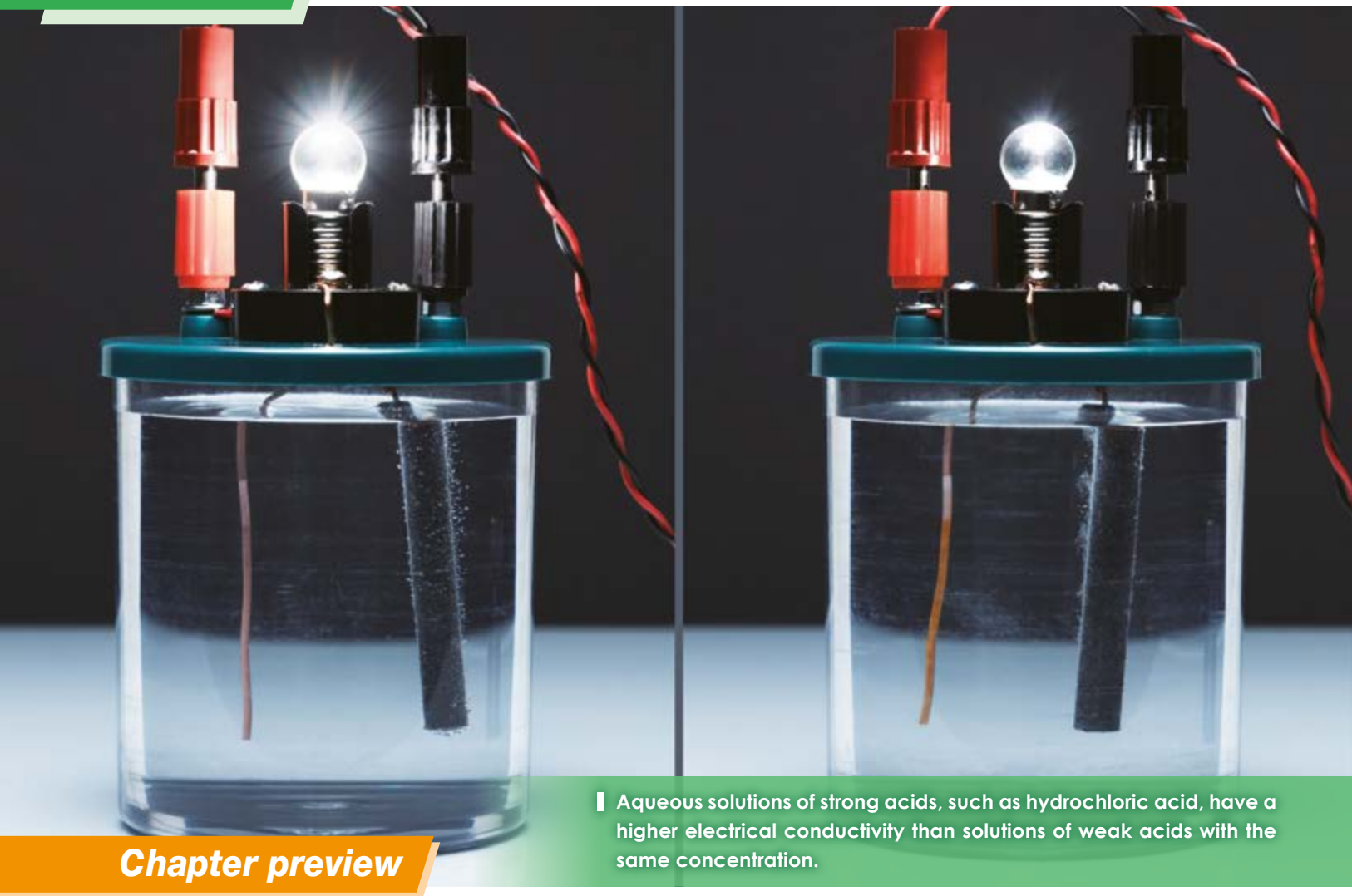
17 Strength of acids and alkalis



Aqueous solutions of strong acids, such as hydrochloric acid, have a higher electrical conductivity than solutions of weak acids with the same concentration.

Chapter preview

17.1 Defining strong and weak acids

17.2 Defining strong and weak alkalis

17.3 Methods to compare the strength of acids/alkalis

17.4 Strength and concentration of acids and alkalis — summary

Prior knowledge & Quick review
(For Chapter 17)

Acids of different strengths

Hydrochloric acid and ethanoic acid both react with magnesium. When magnesium ribbons of equal lengths are added to 1.0 M hydrochloric acid and 1.0 M ethanoic acid respectively, hydrochloric acid reacts more vigorously with the magnesium than ethanoic acid does.



Reactions of 1.0 M hydrochloric acid (left) and 1.0 M ethanoic acid (right) with magnesium ribbons of equal lengths

Think about...

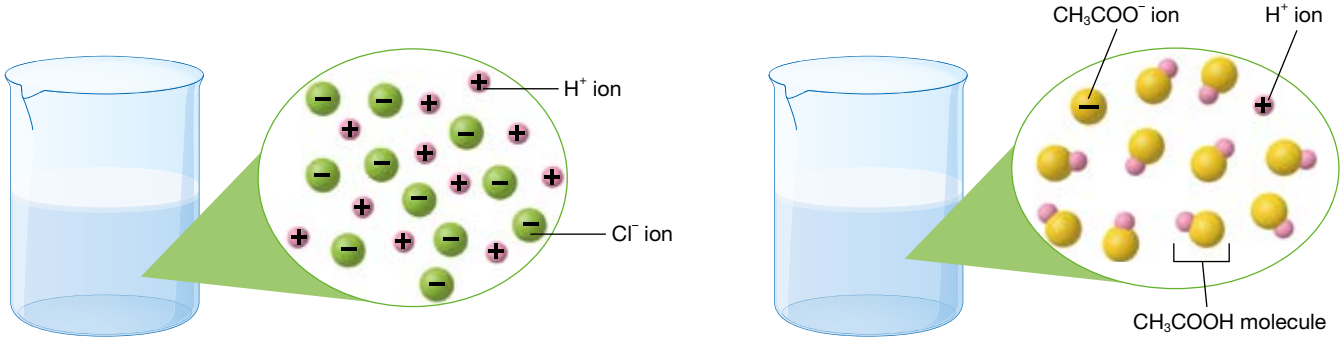
1. What causes the difference in observations in the above reactions?
2. Can you write the equation for the ionization of hydrochloric acid and that of ethanoic acid? How can you account for the above difference in observations based on the equations written?
3. Can you suggest other methods that can compare the strength of different acids?

After studying this chapter, you should be able to answer the above questions.

17.1 Defining strong and weak acids

When hydrogen chloride gas dissolves in water, all the hydrogen chloride molecules **completely ionize** to form hydrogen ions and chloride ions (Figure 17.1(a)).

$$\underset{\text{molecules}}{HCl(g)} \xrightarrow{\text{water}} \underbrace{H^+(aq) + Cl^-(aq)}_{\text{ions}} \quad \textit{complete ionization}$$



(a) 0.1 M solution of hydrochloric acid (b) 0.1 M solution of ethanoic acid

Figure 17.1 (a) A 0.1 M solution of hydrochloric acid contains much more $H^+(aq)$ ions than (b) a 0.1 M solution of ethanoic acid though they are of the same molar concentration. (The diagrams do not show the water molecules present and the relative amounts of ethanoic acid molecules, hydrogen ions and ethanoate ions shown are approximate.)

When pure ethanoic acid dissolves in water, only a small percentage of ethanoic acid molecules ionizes. That is, the acid molecules only **slightly ionize**. In a 0.1 M solution of ethanoic acid, about 13 out of 1000 CH_3COOH molecules ionize to produce $H^+(aq)$ ions.

$$\underset{\text{molecules}}{CH_3COOH(\ell)} \overset{\text{water}}{\rightleftharpoons} \underbrace{H^+(aq) + CH_3COO^-(aq)}_{\text{ions}} \quad \textit{slight ionization}$$

Note that a double arrow '$\rightleftharpoons$' is used in the above equation. This shows that in the solution, ethanoic acid molecules, hydrogen ions and ethanoate ions are all present (Figure 17.1(b)).

Acids can be classified into **strong acids** and **weak acids**. The strength of acids is related to their **extent of ionization** in water. Hydrochloric acid (HCl) is a strong acid while ethanoic acid (CH_3COOH) is a weak acid.

Key point

A **strong acid** is an acid which **ionizes completely (or almost completely)** in water.
A **weak acid** is an acid which **ionizes slightly** in water.

strong acid 強酸
weak acid 弱酸

Table 17.1 shows some common strong acids and weak acids.

Strength	Acid	Ionization in water
Strong	Hydrochloric acid	$HCl(g) \xrightarrow{water} H^+(aq) + Cl^-(aq)$
	Nitric acid	$HNO_3(\ell) \xrightarrow{water} H^+(aq) + NO_3^-(aq)$
	Sulphuric acid	$H_2SO_4(\ell) \xrightarrow{water} H^+(aq) + HSO_4^-(aq)$ $HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$
Weak	Ethanoic acid	$CH_3COOH(\ell) \overset{water}{\rightleftharpoons} H^+(aq) + CH_3COO^-(aq)$
	Carbonic acid	$H_2CO_3(\ell) \overset{water}{\rightleftharpoons} H^+(aq) + HCO_3^-(aq)$ $HCO_3^-(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq)$
	Sulphurous acid	$H_2SO_3(\ell) \overset{water}{\rightleftharpoons} H^+(aq) + HSO_3^-(aq)$ $HSO_3^-(aq) \rightleftharpoons H^+(aq) + SO_3^{2-}(aq)$

Table 17.1 Some common strong acids and weak acids

Learning tip

Most organic acids such as ethanoic acid and citric acid are weak acids.

Class practice 17.1

Write all kinds of particles present in each of the following substances, and mark, with an asterisk, the kind of particles in the greatest amount (apart from water molecules).

(a) Pure sulphuric acid (b) 2.0 M nitric acid

(c) 2.0 M citric acid

17.2 Defining strong and weak alkalis

Similarly, alkalis can be classified into **strong alkalis** and **weak alkalis**. The strength of alkalis is related to their **extent of ionization or dissociation** in water. Sodium hydroxide (NaOH) is a strong alkali. Ammonia (NH_3) is a weak alkali.

Sodium hydroxide is an ionic compound. When sodium hydroxide solid dissolves in water, the closely packed sodium ions and hydroxide ions separate and become mobile. We say that, sodium hydroxide **dissociates** (separates) **completely** in water (Figure 17.2(a)).

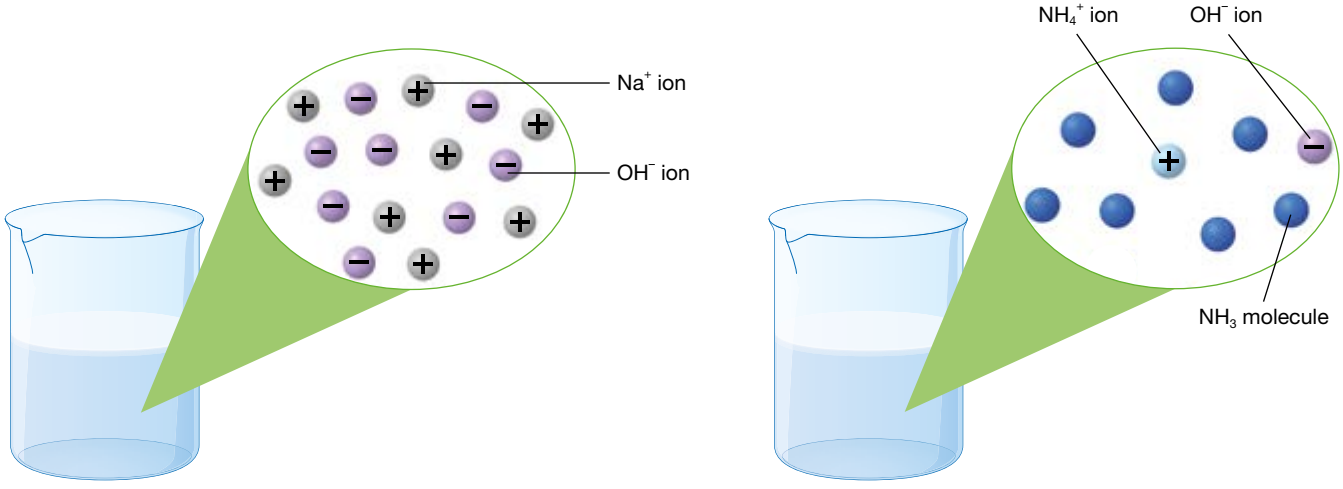
$$\underset{\substack{\text{ions}\\(\textit{not}\ \text{mobile})}}{NaOH(s)} \xrightarrow{water} \underbrace{Na^+(aq) + OH^-(aq)}_{\text{mobile ions}}$$

complete dissociation

organic acid 有機酸
strong alkali 強鹼
weak alkali 弱鹼

Ammonia is a molecular compound. When ammonia gas dissolves in water, only a small percentage of the ammonia molecules ionizes to form ammonium ions and hydroxide ions. That is, it only **slightly ionizes** (Figure 17.2(b)).

$$\underset{\text{molecules}}{NH_3(aq) + H_2O(\ell)} \rightleftharpoons \underbrace{NH_4^+(aq) + OH^-(aq)}_{\text{ions}} \quad \textit{slight ionization}$$



(a) 0.1 M sodium hydroxide solution

(b) 0.1 M aqueous ammonia

Figure 17.2 (a) A 0.1 M sodium hydroxide solution contains much more $OH^-(aq)$ ions than (b) a 0.1 M aqueous ammonia though they are of the same molar concentration. (The diagrams do not show the water molecules present and the relative amounts of ammonia molecules, ammonium ions and hydroxide ions shown are approximate.)

Key point

A **strong alkali** is an alkali which **dissociates completely** in water.
A **weak alkali** is an alkali which **ionizes slightly** in water.

Class practice 17.2

Write all kinds of particles present in each of the following solutions, and mark with an asterisk, the kind of particles in the greatest amount (apart from water molecules).

(a) 1.0 M potassium hydroxide solution

(b) 1.0 M aqueous ammonia

(c) 1.0 M calcium hydroxide solution

17.3 Methods to compare the strength of acids/alkalis

Experiment 17.1

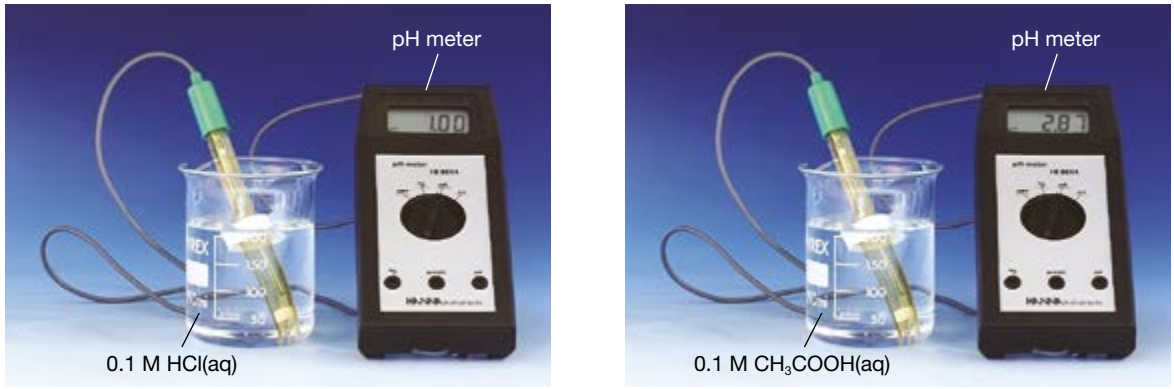
Comparing the strength of acids and alkalis

Experiment video
Comparing the strength of acids and alkalis

When we compare the strength of different acids (or different alkalis), the acids (or alkalis) used should be of the **same molar concentration**.

Measuring pH

We can compare the strength of acids (or alkalis) by measuring their pH using a pH meter. Figure 17.3 shows how we can measure the pH of 0.1 M hydrochloric acid and 0.1 M ethanoic acid.



(a) (b)

Figure 17.3 Set-ups for measuring the pH of (a) 0.1 M hydrochloric acid and (b) 0.1 M ethanoic acid

Figure 17.3 shows that although the hydrochloric acid and the ethanoic acid have the same molar concentration, they have different pH. The pH of the hydrochloric acid is **lower** than that of the ethanoic acid. This is because hydrochloric acid has a **higher concentration of H^+(aq) ions**.

We can perform the above experiment on any set of acids (and alkalis) of different strengths, and the results show that:

- a solution of a **strong acid has a lower pH than that of a weak acid** of the same molar concentration. (There is a **higher concentration of H^+(aq) ions** in the solution of strong acid compared with that of the weak acid.)
- a solution of a **strong alkali has a higher pH than that of a weak alkali** of the same molar concentration. (There is a **higher concentration of OH^-(aq) ions** in the solution of strong alkali compared with that of the weak alkali. The solution of strong alkali is **more alkaline** than that of the weak alkali.)

Example 17.1

Integrated

Comparing the pH of a strong acid solution and a weak acid solution

In an experiment, a student compared the strength of hydrochloric acid and ethanoic acid by measuring the pH of their 0.1 M aqueous solutions. The results were recorded as shown below:

pH of 0.1 M hydrochloric acid = 1.00; pH of 0.1 M ethanoic acid = 2.87

(a) Explain, with the aid of an equation, why the 0.1 M ethanoic acid has a pH higher than 1.00.

(b) From the experimental results, calculate the concentration of $H^+(aq)$ ions in each of the above acids.

(c) Would the pH be lower than, equal to or higher than 1.00 when the 0.1 M hydrochloric acid is replaced by 0.1 M sulphuric acid? Explain your answer.

Solution

(a) Ethanoic acid is a weak acid. It ionizes only slightly in water.

$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

As $[H^+(aq)]$ in 0.1 M $CH_3COOH(aq)$ is smaller than 0.1 M, the pH of the solution will be greater than 1.00.

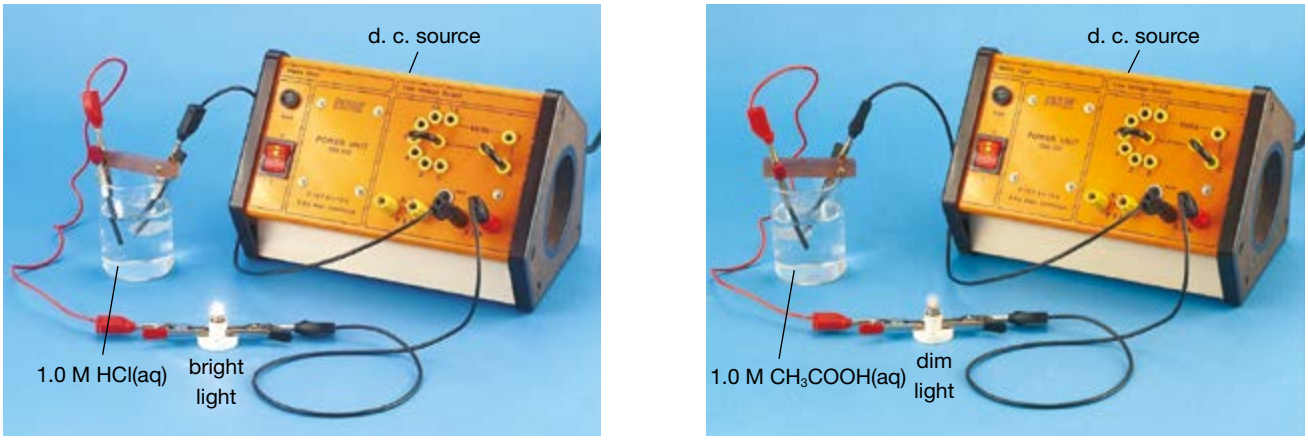
(b) For 0.1 M HCl(aq), $[H^+(aq)] = 10^{-1.00}$ mol dm^{-3} = 0.1 mol dm^{-3}

For 0.1 M $CH_3COOH(aq)$, $[H^+(aq)] = 10^{-2.87}$ mol dm^{-3} = 1.35×10^{-3} mol dm^{-3}

(c) The pH would be lower than 1.00. H_2SO_4 is a dibasic and strong acid. For the same molar concentration, $H_2SO_4(aq)$ has a higher concentration of $H^+(aq)$ ions than HCl(aq).

Measuring electrical conductivity

We can also compare the strength of acids (or alkalis) by measuring the electrical conductivity of their aqueous solutions. Figure 17.4 shows how we can measure the electrical conductivities of 1.0 M hydrochloric acid and 1.0 M ethanoic acid.



(a) (b)

Figure 17.4 Set-ups for comparing the electrical conductivity of (a) 1.0 M hydrochloric acid and (b) 1.0 M ethanoic acid

Learning tip

The electrical conductivity of a solution increases with the concentration of the mobile ions present.

In Figure 17.4, the light bulb in the set-up for hydrochloric acid glows brighter than that for ethanoic acid. This shows that hydrochloric acid has a **higher electrical conductivity** than ethanoic acid of the same molar concentration, because hydrochloric acid contains a higher concentration of mobile ions ($H^+(aq)$ and $Cl^-(aq)$).

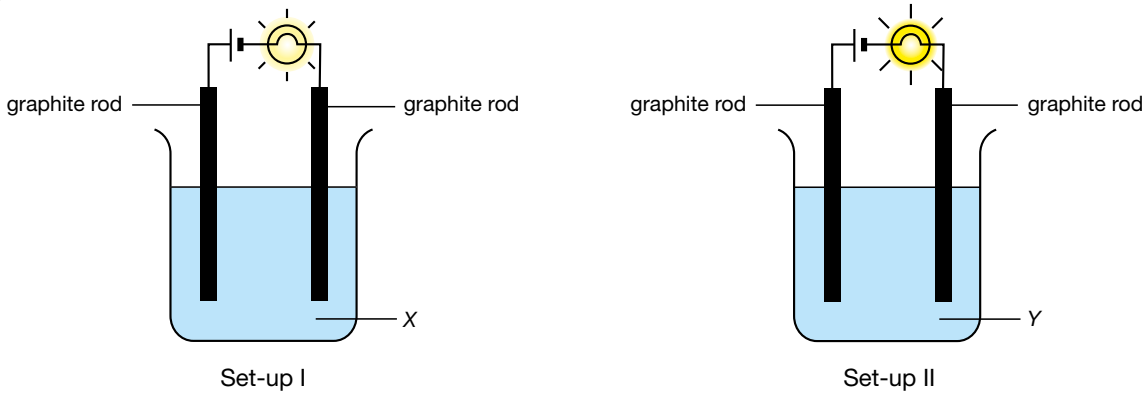
We can perform the above experiment on any set of acids (and alkalis) of different strengths, and the results show that:

- a solution of a **strong acid conducts much better than that of a weak acid** of the same molar concentration. (There is a **higher concentration of mobile ions** in the solution of strong acid compared with that of the weak acid.)
- a solution of a **strong alkali conducts much better than that of a weak alkali** of the same molar concentration. (There is a **higher concentration of mobile ions** in the solution of strong alkali compared with that of the weak alkali.)

Example 17.2

Comparing the electrical conductivities of a strong alkali and a weak alkali

A student was given two solutions labelled *X* and *Y*, respectively. One of which was 1.0 M sodium hydroxide solution and the other one was 1.0 M aqueous ammonia. The student tried to distinguish the two solutions by measuring their electrical conductivities using the set-ups shown in the diagram below.



The student found that the light bulb in Set-up II glowed brighter than that in Set-up I.

(a) From the experiment results, identify *X* and *Y*.

(b) Explain your answer in (a) with the aid of equations.

Solution

(a) *X* was 1.0 M $NH_3(aq)$ while *Y* was 1.0 M NaOH(aq).

(b) NaOH(aq) is a stronger alkali than $NH_3(aq)$. NaOH(aq) dissociates completely in water while $NH_3(aq)$ ionizes only slightly in water.

Cont'd

$NaOH(aq) \longrightarrow Na^+(aq) + OH^-(aq)$

$NH_3(aq) + H_2O(\ell) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

The concentration of mobile ions in 1.0 M NaOH(aq) is higher compared with 1.0 M NH_3(aq). Hence, 1.0 M NaOH(aq) has a higher electrical conductivity. This explains why the light bulb in Set-up II glowed brighter.

Example 17.3

Distinguishing a strong acid from a weak acid with the same pH

You are provided with two solutions, one of which is hydrochloric acid and the other is ethanoic acid. Both solutions have a pH of 3.0.

(a) Suggest a physical method to distinguish the two solutions.

(b) Explain your method suggested in (a).

(c) A student proposed that the two solutions can be distinguished from each other by adding two identical magnesium ribbons to the acids respectively. This is because the stronger acid will give off hydrogen gas bubbles at a higher rate. Do you agree with the student? Explain your answer.

Solution

(a) Dilute each of the two acid solutions by 10 times by adding distilled water. Then measure their pH by using a pH meter. The pH of the diluted ethanoic acid will be lower than that of the diluted hydrochloric acid.

(b) As hydrochloric acid is a strong acid, it ionizes completely in water. When it is diluted by 10 times, the concentration of $H^+(aq)$ ions will decrease by 10 times (i.e. the concentration will be 10^{-4} mol dm^{-3}). Hence, the pH of the resultant solution is 4.0.

Ethanoic acid is a weak acid. It ionizes incompletely in water. When it is diluted by 10 times, besides the dilution of the $H^+(aq)$ ions originally present, some of the unionized ethanoic acid will ionize to give additional $H^+(aq)$ ions.

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

Thus, the concentration of $H^+(aq)$ ions in the resultant solution will be greater than 10^{-4} mol dm^{-3} and the diluted solution has a pH smaller than 4.0.

(c) No, the two solutions will give off hydrogen at the same rate instantaneously because both solutions contain $H^+(aq)$ ions at the same concentration, i.e. 10^{-3} mol dm^{-3}.

Experiment 17.2

Distinguishing a strong acid and a weak acid having the same pH

Experiment video

Distinguishing a strong acid and a weak acid having the same pH

There are other methods used to compare the strength of acids/alkalis. Some involve the chemical reactions of the acids/alkalis. Whatever method is used, it is essential that the comparison is a fair one. For example, we can compare the strength of acids by studying their reactions with a metal (refer to the Chapter opening on p.92), a solid carbonate or a solid hydrogencarbonate.

17.4 Strength and concentration of acids and alkalis — summary

We should not mix up the terms **strength** (described by the words **strong** or **weak**) and **concentration** (described by the words **concentrated** or **dilute**).

- **Strength** refers to the **extent of ionization or dissociation** of an acid or alkali **in water**.
- **Concentration** refers to the **amount** of an acid or alkali **per unit volume of the solution**.

Figure 17.5 illustrates the differences between strength and concentration using a monobasic strong acid (HCl) and a monobasic weak acid (CH_3COOH) as examples.

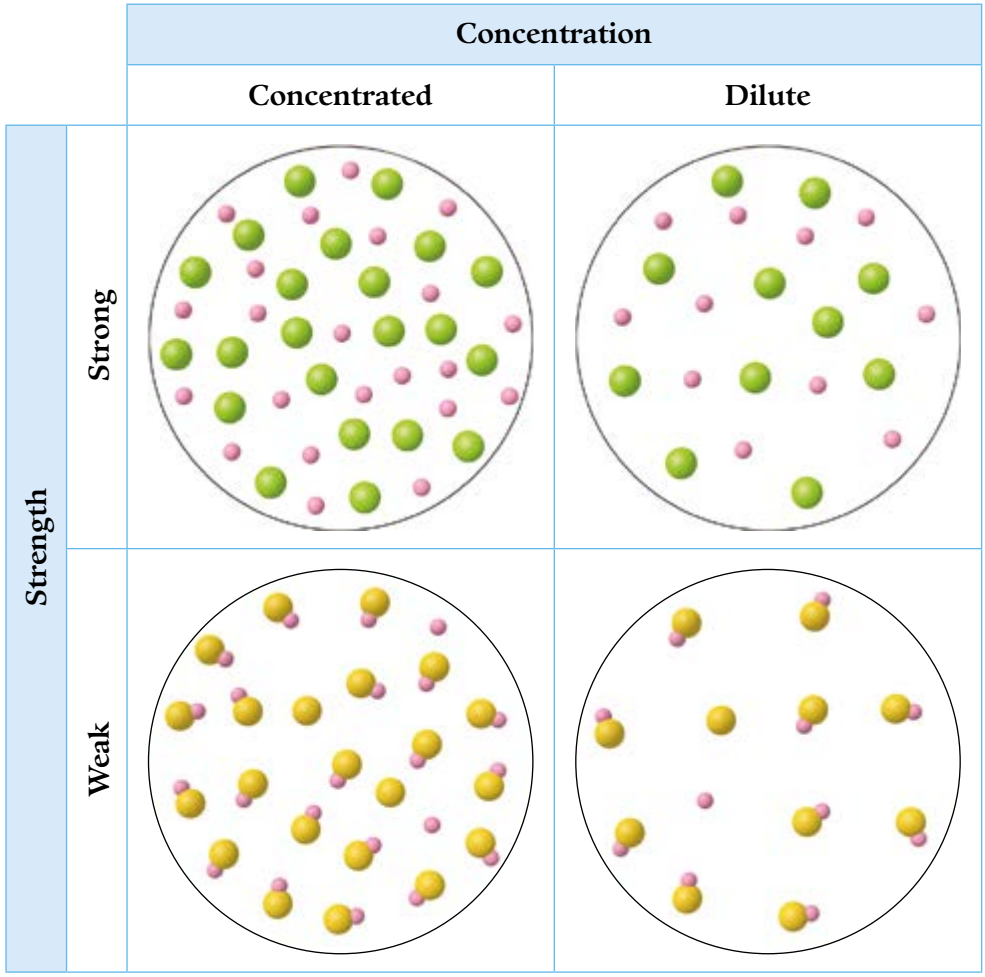


Figure 17.5 A diagram illustrating the difference between strength and concentration of acids (using monobasic acids, HCl and CH_3COOH, as examples). (Note: the diagrams do not show the water molecules present.)

The following examples further illustrate the concepts of strength and concentration.

Aqueous solution of acid or alkali	Concentration and strength
1 M HCl(aq)	a dilute solution of a strong acid
12 M HCl(aq)	a concentrated solution of a strong acid
1 M CH_3COOH(aq)	a dilute solution of a weak acid
12 M CH_3COOH(aq)	a concentrated solution of a weak acid
1 M NaOH(aq)	a dilute solution of a strong alkali
10 M NH_3(aq)	a concentrated solution of a weak alkali

In conclusion, the strength of an acid or alkali is not related to its concentration. An aqueous solution of a strong acid or a strong alkali can be a concentrated or dilute solution. An aqueous solution of a weak acid or a weak alkali can also be a concentrated or dilute solution.

Class practice 17.3

1. The pH of 0.1 M sodium hydroxide solution and that of 0.1 M aqueous ammonia are shown below.

	0.1 M sodium hydroxide solution	0.1 M aqueous ammonia
pH	13.00	11.12

   (a) What ion is responsible for the alkaline properties of the two solutions?

   (b) Write the equation for the formation of the ion mentioned in (a) for aqueous ammonia.

   (c) With reference to the pH values, state which alkali is stronger. Explain your answer.

2. A student was given two beakers of dilute solutions of alkalis *P* and *Q*. Then the student measured the pH of these two solutions. It was found that the solution of alkali *P* had a higher pH than the solution of alkali *Q*. Which of the following deductions about *P* and *Q* is correct?

   A. *P* is a stronger alkali than *Q*.
   B. *Q* is a stronger alkali than *P*.
   C. *P* has a higher concentration of OH^-(aq) ions than *Q*.
   D. *Q* has a higher concentration of OH^-(aq) ions than *P*. ☐

3. You are provided with three acid solutions: 2.0 M hydrochloric acid, 2.0 M ethanoic acid and 2.0 M sulphuric acid.

   (a) Suggest a method for measuring the pH of these solutions accurately.

   (b) Arrange the three solutions in order of decreasing pH.

Chapter review

Progress check

After studying this chapter, you should be able to:

- ☑ describe the ionization or dissociation of acids and alkalis
- ☑ relate the strength of acids and alkalis to their extent of ionization or dissociation
- ☑ compare the strength of acids and alkalis by the measurement of pH and electrical conductivities
- ☑ describe acids and alkalis with the appropriate terms: strong and weak, concentrated and dilute

Key terms

e-Dictionary (For Chapter 17)

		Page			Page
1.	strong acid 強酸	93	3.	weak acid 弱酸	93
2.	strong alkali 強鹼	94	4.	weak alkali 弱鹼	94

Key concepts

17.1 Defining strong and weak acids

1. A **strong acid** is an acid which ionizes completely (or almost completely) in water e.g. hydrochloric acid, nitric acid and sulphuric acid.

2. A **weak acid** is an acid which ionizes slightly in water e.g. ethanoic acid, carbonic acid and sulphurous acid.

17.2 Defining strong and weak alkalis

3. A **strong alkali** is an alkali which dissociates completely in water e.g. potassium hydroxide and sodium hydroxide.

4. A **weak alkali** is an alkali which ionizes slightly in water e.g. ammonia.

17.3 Methods for comparing the strength of acids/alkalis

5. Strength of acids or alkalis can be compared by measuring the pH, and electrical conductivity of their aqueous solutions of the same molarity.

17.4 Strength and concentration of acids and alkalis — summary

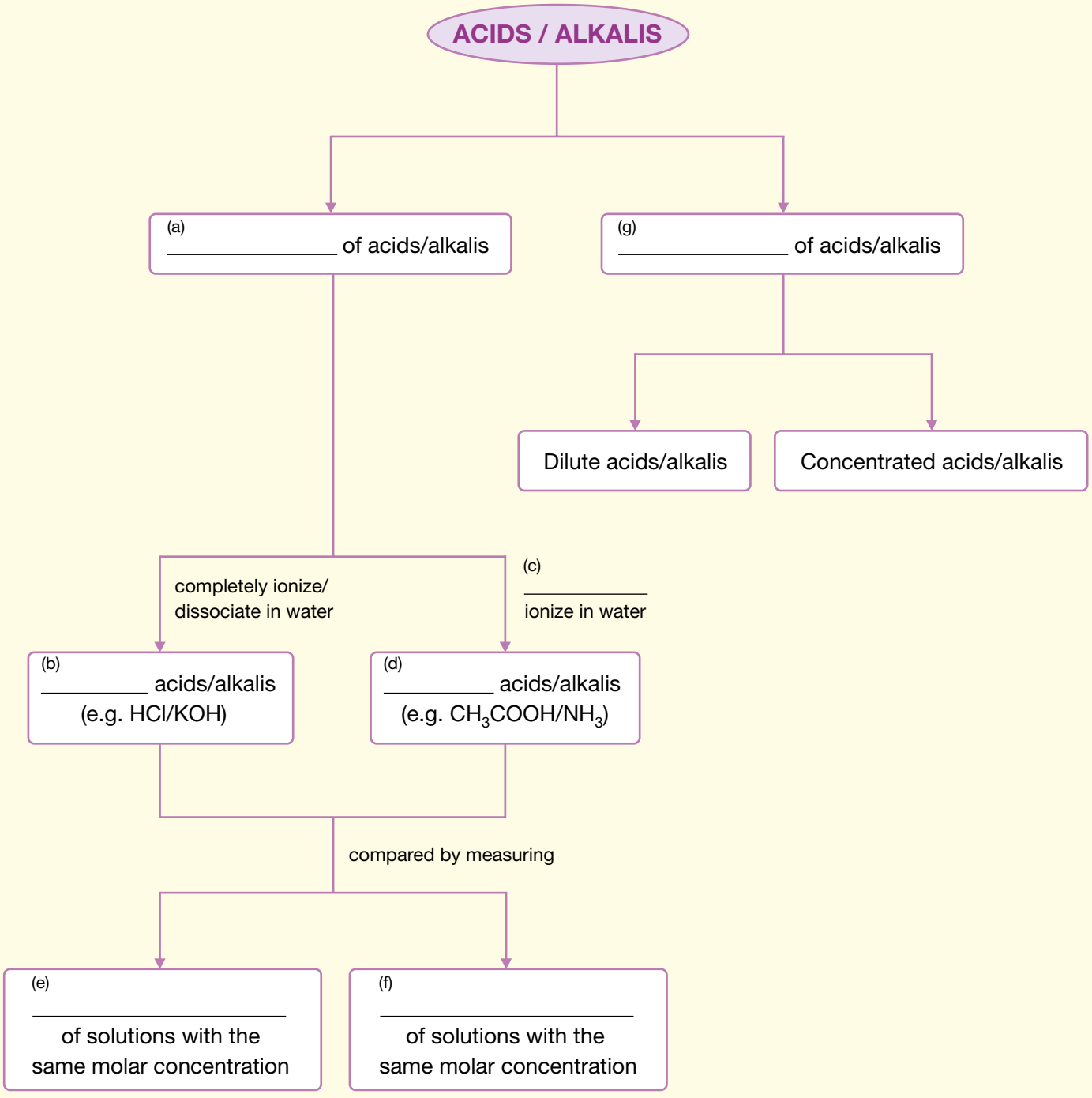
6. The strength of an acid or an alkali refers to the extent of ionization or dissociation of the acid or the alkali in water.

7. The concentration of an acid or an alkali refers to the amount of the acid or the alkali per unit volume of the solution.

Concept map

Complete the concept map by filling in the items listed below:

concentration, electrical conductivities, pH, slightly, strength, strong, weak



Chapter exercise

A. Fill in the blanks

Section 17.1

1. (a) A strong acid is an acid which ionizes _____ (or _____ _____) in water. An example is _____ .

   (b) A weak acid is an acid which ionizes _____ in water. An example is _____ .

Section 17.2

2. (a) A strong alkali is an alkali which _____ completely in water. An example is _____ .

   (b) A weak alkali is an alkali which ionizes _____ in water. An example is _____ .

Section 17.3

3. For solutions of the same molar concentration,

   (a) the electrical conductivity of a solution of a strong acid/alkali is _____ than that of a weak acid/alkali;

   (b) the pH of a solution of a strong acid is _____ than that of a weak acid; and

   (c) the pH of a solution of a strong alkali is _____ than that of a weak alkali.

Section 17.4

4. (a) 0.5 M hydrochloric acid is a _____ solution (in terms of concentration) of _____ acid (in terms of strength).

   (b) 8.0 M ethanoic acid is a _____ solution (in terms of concentration) of _____ acid (in terms of strength).

   (c) 10.0 M sodium hydroxide solution is a _____ solution (in terms of concentration) of _____ alkali (in terms of strength).

   (d) 0.5 M aqueous ammonia is a _____ solution (in terms of concentration) of _____ alkali (in terms of strength).

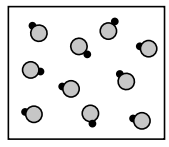
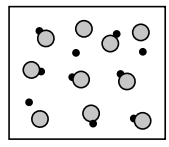
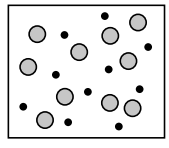
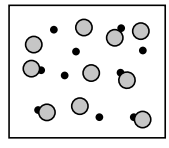
B. Multiple-choice questions

Section 17.1

5. Hydrochloric acid is a strong acid because it

   A. reacts with strong alkalis.
   B. is corrosive.
   C. dissolves in water completely.
   D. ionizes completely in water. ☐

6. Which of the following diagrams best represents an aqueous solution of strong acid?

   A. 
   B. 
   C. 
   D.  ☐

   (• represents $H^+(aq)$, ○ represents the anion of the acid after ionization; water molecules are not shown.)

Section 17.2

7. Which of the following particles are present in 1.0 M aqueous ammonia?

   (1) $NH_3(aq)$ (2) $NH_4^+(aq)$
   (3) $H_2O(\ell)$

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

Section 17.4

8. Which of the following acids are strong acids?

   (1) 0.1 M $H_2SO_4(aq)$
   (2) 5.0 M $CH_3COOH(aq)$
   (3) 14.0 M $HNO_3(aq)$

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

Section 17.3

9. Consider the following statements and choose the best answer:

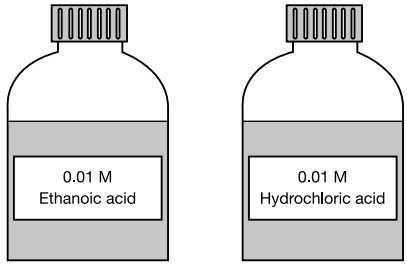
1st statement	2nd statement
Copper does not react with dilute ethanoic acid.	Dilute ethanoic acid is a weak acid.

   A. Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
   B. Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
   C. The 1st statement is false but the 2nd statement is true.
   D. Both statements are false. ☐

C. Structured questions

Section 17.3

10. Consider the following two acids.



   (a) Calculate the pH of 0.01 M hydrochloric acid. (1 mark)

   (b) Would you expect the pH of 0.01 M ethanoic acid to be lower, equal to or higher than that of 0.01 M hydrochloric acid? Explain your answer. (2 marks)

   (c) Which of these acids has a higher electrical conductivity? Explain your answer. (2 marks)

   (d) When two identical iron nails are put into the acids separately, predict which acid will react more vigorously with the iron nail. Explain briefly. (2 marks)

11. The pH of a sample of ethanoic acid is 3.0.

   (a) What is the concentration of $H^+(aq)$ ions in the sample of ethanoic acid? (1 mark)

   (b) Explain whether the concentration of the ethanoic acid is equal to the concentration of $H^+(aq)$ ions calculated in (a). (1 mark)

Exam practice

A. Multiple-choice questions

Section 17.1

1. Which of the following statements about 1.0 M sulphurous acid are correct?

   (1) It ionizes slightly in water to give $H^+(aq)$.
   (2) It contains $HSO_3^-(aq)$ ions.
   (3) The concentration of $H^+(aq)$ ions in the acid is 2.0 M.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3)

2. Which of the following statements concerning nitric acid are correct?

   (1) Nitric acid is a mineral acid.
   (2) Dilute nitric acid contains $H^+(aq)$ ions, $NO_3^-(aq)$ ions and $OH^-(aq)$ ions.
   (3) Dilute nitric acid reacts with sodium carbonate to give a colourless gas.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3)

Section 17.3

3. Which of the following combinations about 1.0 M NaOH(aq) and 1.0 M $NH_3(aq)$ is INCORRECT?

	1.0 M NaOH(aq)	1.0 M $NH_3(aq)$
A.	Dissociates completely in water	Ionizes slightly in water
B.	Has a higher concentration of $OH^-(aq)$ ions	Has a lower concentration of $OH^-(aq)$ ions
C.	Forms a precipitate with $Mg(NO_3)_2(aq)$	Does not form a precipitate with $Mg(NO_3)_2(aq)$
D.	Has a higher electrical conductivity	Has a lower electrical conductivity

4. Which of the following statements about carbonic acid and sulphuric acid of the same molarity is correct?

   A. The carbonic acid has a lower pH than the sulphuric acid.
   B. The carbonic acid has a lower electrical conductivity than the sulphuric acid.
   C. The sulphuric acid has a lower concentration of $H^+(aq)$ ions than the carbonic acid.
   D. The sulphuric acid reacts with sodium hydrogencarbonate slower than the carbonic acid.

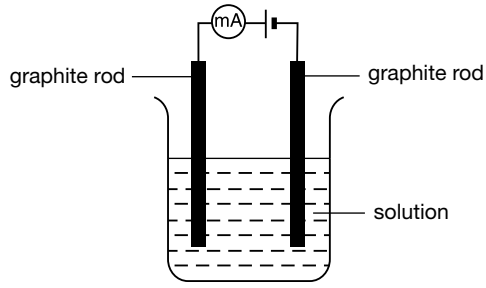
5. Consider the following acids listed below:

   (1) 1.0 M hydrochloric acid
   (2) 1.0 M sulphuric acid
   (3) 1.0 M ethanoic acid

   Which of the following represents the acids arranged in order of increasing acidity?

   A. (2), (1), (3)
   B. (2), (3), (1)
   C. (3), (1), (2)
   D. (3), (2), (1)

6. Refer to the set-up shown in the diagram below.



   Which of the following solutions would give the smallest current reading?

   A. 1.0 M aqueous ammonia
   B. 1.0 M hydrochloric acid
   C. 1.0 M sodium chloride solution
   D. 1.0 M sodium hydroxide solution

7. The pH of a sample of aqueous ammonia and that of a sample of sodium hydroxide solution are both 10.0. Which of the following statements about these samples is/are correct?

(1) They have the same concentration of $OH^-(aq)$ ions.
(2) They are of the same concentration.
(3) They are of the same strength.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

Section 17.4

8. Which of the following solutions has the highest pH?

A. 0.010 M aqueous ammonia
B. 0.100 M aqueous ammonia
C. 0.010 M potassium hydroxide solution
D. 0.100 M potassium hydroxide solution

9. Consider a solution of a weak acid with a concentration of 0.01 mol dm^{-3}. Which of the following combinations about the electrical conductivity and the pH of this solution is correct?

	Electrical conductivity	pH
A.	Good	12.5
B.	Poor	7.0
C.	Poor	3.5
D.	Good	2.0

10. Refer to the information in the table below:

Solution	Contents	pH
X	50 cm^3 of 0.001 M HCl(aq)	3.0
Y	25 cm^3 of 0.001 M H_2SO_4(aq)	2.7
Z	50 cm^3 of 0.1 M CH_3COOH(aq)	2.9

Which of the following statements is correct?

A. *X* has a higher pH than *Z* because HCl is a stronger acid than CH_3COOH.
B. *Y* has a lower pH than *X* because the volume of H_2SO_4(aq) is smaller than that of HCl(aq).
C. *Y* has a lower pH than *X* because H_2SO_4 is a strong dibasic acid but HCl is a strong monobasic acid.
D. *Y* has a lower pH than *Z* because the concentration of H_2SO_4(aq) is lower than that of CH_3COOH(aq).

(HKDSE 2021 Paper 1A Q6)

B. Structured questions

Section 17.3

11. A student compared the strengths of three monobasic acids, *X*, *Y* and *Z* by measuring the pH of their aqueous solutions of the same molarity. The results are tabulated.

Acid	pH
X	4.0
Y	2.0
Z	5.0

(a) *Y* is known to be a strong acid. Calculate the molarity of *Y* in the solution. (1 mark)

(b) Compare the strengths of *X* and *Z*. Explain your answer. (2 marks)

Exam practice

12. Ammonia is an alkali as it produces hydroxide ions when dissolved in water.

(a) Write the equation for the ionization of ammonia in water. (1 mark)

(b) Outline an experiment that can compare the strengths of aqueous ammonia and sodium hydroxide solution. You should include in your answer the expected results. (3 marks)

13. The strength of an acid can be reflected from its reaction with a solid carbonate.

(a) You are provided with lumps of magnesium carbonate. Outline an experiment to show that sulphuric acid is a stronger acid than sulphurous acid. (2 marks)

(b) State and explain the expected observations in (a). (2 marks)

Answers to Chapter exercise

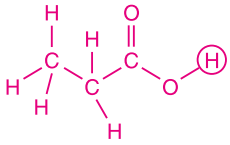
Chapter 14

A. Fill in the blanks (p.41)

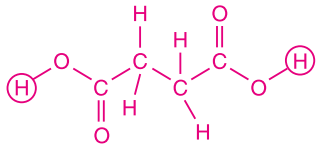
1. hydrogen ions/$H^+(aq)$
2. water; water; hydrogen
3. basicity
4. salt; water; alkali
5. hydroxide; $OH^-(aq)$
6. corrosive
7. hydrogen ions/$H^+(aq)$; oxidizing; oxidizing; dehydrating

B. Practice questions (p.41)

8. (a) For propanoic acid:



For butanedioic acid:



(b) Basicity of propanoic acid is 1./Propanoic acid is a monobasic acid.
Basicity of butanedioic acid is 2./Butanedioic acid is a dibasic acid.

9. (a) Iron powder reacts slowly with the acid and dissolves (to give a pale green solution). Colourless gas bubbles form.
$Fe(s) + 2H^+(aq) \longrightarrow Fe^{2+}(aq) + H_2(g)$

(b) Solid magnesium oxide dissolves.
$MgO(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2O(l)$

(c) The solid magnesium hydroxide dissolves.
$Mg(OH)_2(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + 2H_2O(l)$

(d) Colourless gas bubbles form/effervescene occurs. Sodium hydrogencarbonate powder dissolves.
$HCO_3^-(aq) + H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$

(e) Colourless gas bubbles form/effervescene occurs. Sodium carbonate powder dissolves.
$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$

(f) A colourless gas with a pungent smell evolves and the gas can turn moist red litmus paper blue.
$NH_4^+(aq) + OH^-(aq) \longrightarrow NH_3(g) + H_2O(l)$

(g) A white precipitate forms. The white precipitate dissolves to give a colourless solution when excess sodium hydroxide solution is added.
$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
$Pb(OH)_2(s) + 2OH^-(aq) \longrightarrow Pb(OH)_4^{2-}(aq)$

(h) A white precipitate forms. The white precipitate dissolves to give a colourless solution when excess aqueous ammonia is added.
$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$
$Zn(OH)_2(s) + 4NH_3(aq) \longrightarrow Zn(NH_3)_4^{2+}(aq) + 2OH^-(aq)$

10. (a) Add sodium hydroxide solution to each of the solutions until in excess. A white precipitate forms initially in each solution. But only the precipitate of aluminium nitrate dissolves in excess sodium hydroxide solution.

(b) Add dilute hydrochloric acid to each of the solutions respectively. Potassium carbonate solution gives out colourless gas bubbles while there is no observable change for potassium hydroxide solution.

C. Multiple-choice questions (p.42)

11. C
Tea contains tannic acid.
12. A
A 'pop' sound is heard when the hydrogen formed is tested by a burning splint.
13. B
14. B
15. A
Iron(III) hydroxide is not soluble in water. It is a base but not an alkali.
16. A
(1): $2KOH(aq) + SO_2(g) \longrightarrow K_2SO_3(aq) + H_2O(l)$
(2): $2KOH(aq) + CO_2(g) \longrightarrow K_2CO_3(aq) + H_2O(l)$
17. D
Sodium hydroxide solution has no reaction with zinc metal. Dilute hydrochloric acid has no reaction with ammonium chloride solution.
18. B
(2): a white precipitate of $Pb(OH)_2$ forms.
(3): a dark brown precipitate of Ag_2O forms when a small amount of $NH_3(aq)$ is added. The precipitate dissolves in excess $NH_3(aq)$ to form a colourless solution.

19. D

$(NH_4)_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NH_4Cl(aq) + CO_2(g) + H_2O(l)$

The carbon dioxide evolved can turn limewater milky.

The ammonia evolved can turn moist red litmus paper blue.

20. D
21. C
22. B
23. C

D. Structured questions (p.44)

24. (a) Citrus fruits/oranges/lemons/grapefruits/tomatoes/broccoli/potatoes 1

(b) Carbon dioxide 1

G can be tested by using limewater. 1

It turns limewater milky. 1

(c) When the vitamin C tablet is added to water, the citric acid in the tablet will dissolve and ionize in water to give $H^+(aq)$, which reacts with the sodium hydrogencarbonate/$HCO_3^-(aq)$ to give carbon dioxide. 1

$HCO_3^-(aq) + H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$ 1

25. (a) $Fe^{2+}(aq)$ and $Cu^{2+}(aq)$ 1

Colour of $Fe(OH)_2(s)$: dirty green } 1

Colour of $Cu(OH)_2(s)$: pale blue }

(b) $K^+(aq)$ 1

(c) $Mg^{2+}(aq)$

$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$ 1

Chapter 15

A. Fill in the blanks (p.65)

1. amount of solute
2. mass; volume
3. moles; volume
4. molar mass
5. remains unchanged; M_2V_2

B. Practice questions (p.65)

6. (a) (i) Concentration of $MgSO_4(aq)$

$= \dfrac{3.01}{\frac{250.0}{1000}}\ g\ dm^{-3} = 12.0\ g\ dm^{-3}$

(ii) Molar mass of $MgSO_4$

$= (24.3 + 32.1 + 16.0 \times 4)\ g\ mol^{-1}$

$= 120.4\ g\ mol^{-1}$

Molarity of $MgSO_4(aq)$

$= \dfrac{12.0}{120.4}\ mol\ dm^{-3} = 0.0997\ mol\ dm^{-3}$

(b) (i) Concentration of $NH_4Cl(aq)$

$= \dfrac{2.14}{\frac{50.0}{1000}}\ g\ dm^{-3} = 42.8\ g\ dm^{-3}$

(ii) Molar mass of NH_4Cl

$= (14.0 + 1.0 \times 4 + 35.5)\ g\ mol^{-1}$

$= 53.5\ g\ mol^{-1}$

Molarity of $NH_4Cl(aq)$

$= \dfrac{42.8}{53.5}\ mol\ dm^{-3} = 0.800\ mol\ dm^{-3}$

(c) (i) Concentration of $Ca(NO_3)_2(aq)$

$= \dfrac{6.47}{\frac{100.0}{1000}}\ g\ dm^{-3} = 64.7\ g\ dm^{-3}$

(ii) Molar mass of $Ca(NO_3)_2$

$= [40.1 + (14.0 + 16.0 \times 3) \times 2]\ g\ mol^{-1}$

$= 164.1\ g\ mol^{-1}$

Molarity of $Ca(NO_3)_2(aq)$

$= \dfrac{64.7}{164.1}\ mol\ dm^{-3} = 0.394\ mol\ dm^{-3}$

(d) (i) Concentration of NaCl(aq)

$= \dfrac{8.78}{\frac{500.0}{1000}}\ g\ dm^{-3} = 17.6\ g\ dm^{-3}$

(ii) Molar mass of NaCl

$= (23.0 + 35.5)\ g\ mol^{-1}$

$= 58.5\ g\ mol^{-1}$

Molarity of NaCl(aq)

$= \dfrac{17.6}{58.5}\ mol\ dm^{-3} = 0.301\ mol\ dm^{-3}$

(e) (i) Concentration of $FeSO_4(aq)$

$= \dfrac{7.94}{2.0}\ g\ dm^{-3} = 3.97\ g\ dm^{-3}$

(ii) Molar mass of $FeSO_4$

$= (55.8 + 32.1 + 16.0 \times 4)\ g\ mol^{-1}$

$= 151.9\ g\ mol^{-1}$

Molarity of $FeSO_4(aq)$

$= \dfrac{3.97}{151.9}\ mol\ dm^{-3} = 0.0261\ mol\ dm^{-3}$

7.

Formula of solute	Molar mass / g mol^{-1}	Mass of solute / g	Volume of solution	Molarity			Concentration of solution / g dm^{-3}
				solution	cation	anion	
(a) NaCl	58.5	117	2.00 dm^3	1.00 M	1.00 M	1.00 M	58.5
(b) $AgNO_3$	169.9	17.0	1.00 dm^3	0.100 M	0.100 M	0.100 M	17.0
(c) Na_2CO_3	106.0	2.65	500.0 cm^3	0.0500 M	0.100 M	0.0500 M	5.30
(d) $MgCl_2$	95.3	29.3	1.50 dm^3	0.205 M	0.205 M	0.410 M	19.5
(e) $C_6H_{12}O_6$	180.0	0.346	12.0 cm^3	0.160 M	–	–	28.8

C. Multiple-choice questions (p.66)

8. D

Concentration of K_2SO_4(aq)

$= 0.35 \text{ mol dm}^{-3} \times (39.1 \times 2 + 32.1 + 16.0 \times 4) \text{ g mol}^{-1}$

$= 61.0 \text{ g dm}^{-3}$

9. D

Concentration of NaOH(aq)

$= \dfrac{9.32}{\frac{100.0}{1000}} \text{ g dm}^{-3} = 93.2 \text{ g dm}^{-3}$

Molar mass of NaOH

$= (23.0 + 16.0 + 1.0) \text{ g mol}^{-1} = 40.0 \text{ g mol}^{-1}$

Molarity of NaOH(aq)

$= \dfrac{93.2}{40.0} \text{ mol dm}^{-3} = 2.33 \text{ mol dm}^{-3}$

10. A

Mass of H_2SO_4 present in 1000 cm^3 of the acid

$= 1.20 \text{ g cm}^{-3} \times 1000 \text{ cm}^3 \times 30.0\% = 360 \text{ g}$

Molarity of the acid

$= \dfrac{360 \text{ g dm}^{-3}}{(1.0 \times 2 + 32.1 + 16.0 \times 4) \text{ g mol}^{-1}} = 3.67 \text{ mol dm}^{-3}$

11. B

Number of moles of KNO_3

$= 0.096 \times \dfrac{25.0}{1000} \text{ mol} = 2.4 \times 10^{-3} \text{ mol}$

12. C

Number of moles of NaOH in 250.0 cm^3 of 1.0 M NaOH(aq)

$= 1.0 \times \dfrac{250.0}{1000} \text{ mol} = 0.25 \text{ mol}$

Option (A):

Number of moles of NaOH in 100.0 cm^3 of 4.0 M NaOH(aq)

$= 4.0 \times \dfrac{100.0}{1000} \text{ mol} = 0.40 \text{ mol}$

Option (B):

Number of moles of NaOH in 200.0 cm^3 of 1.5 M NaOH(aq)

$= 1.5 \times \dfrac{200.0}{1000} \text{ mol} = 0.30 \text{ mol}$

Option (C):

Number of moles of NaOH in 500.0 cm^3 of 0.5 M NaOH(aq)

$= 0.5 \times \dfrac{500.0}{1000} \text{ mol} = 0.25 \text{ mol}$

Option (D):

Number of moles of NaOH in 1000 cm^3 of 0.2 M NaOH(aq)

$= 0.2 \times \dfrac{1000}{1000} \text{ mol} = 0.2 \text{ mol}$

13. B

Number of moles of NaOH

$= \dfrac{12.0}{23.0 + 16.0 + 1.0} \text{ mol} = 0.300 \text{ mol}$

Volume of NaOH(aq)

$= \dfrac{0.300}{2.00} \text{ dm}^3 = 0.150 \text{ dm}^3 = 150 \text{ cm}^3$

14. D

Number of moles of $Na_2S_2O_3$

$= 1.50 \times 3.50 \text{ mol} = 5.25 \text{ mol}$

Mass of $Na_2S_2O_3 \cdot 5H_2O$

$= 5.25 \times [(23.0 \times 2 + 32.1 \times 2 + 16.0 \times 3) + (1.0 \times 2 + 16.0) \times 5] \text{ g}$

$= 1303 \text{ g}$

15. C

There are 2 Fe^{3+} ions and 3 SO_4^{2-} ions in each formula unit of $Fe_2(SO_4)_3$.

Hence, 0.13 M of $Fe_2(SO_4)_3$ contains 0.13×2 M = 0.26 M Fe^{3+} ions and $0.13 \times 3 = 0.39$ M SO_4^{2-} ions.

16. B

Number of moles of NaOH in 100.0 cm^3 of 0.5 M NaOH(aq)

$= 0.5 \times \dfrac{100.0}{1000} \text{ mol} = 0.05 \text{ mol}$

Number of moles of NaOH in 200.0 cm^3 of 1.5 M NaOH(aq)

$= 1.5 \times \dfrac{200.0}{1000} \text{ mol} = 0.30 \text{ mol}$

Concentration of the resultant NaOH(aq)

$= \dfrac{(0.05 + 0.30) \text{ mol}}{\frac{100.0 + 200.0}{1000} \text{ dm}^3} = 1.17 \text{ mol dm}^{-3}$

17. C

Number of moles of K_2SO_4

$= 1.0 \times \dfrac{200.0}{1000} \text{ mol} = 0.20 \text{ mol}$

Number of moles of K^+ from K_2SO_4

$= 0.20 \text{ mol} \times 2 = 0.40 \text{ mol}$

Number of moles of KCl

$= 2.0 \times \dfrac{100.0}{1000} \text{ mol} = 0.20 \text{ mol}$

Number of moles of K^+ in the resultant solution
= (0.40 + 0.20) mol = 0.60 mol
Volume of the resultant solution
= (200.0 + 100.0) cm^3 = 300.0 cm^3
Concentration of $K^+(aq)$ in the resultant solution
$= \frac{0.60 \text{ mol}}{\frac{300.0}{1000} \text{ dm}^3} = 2.0 \text{ mol dm}^{-3}$

18. A
During dilution, as only water is added to the solution, the number of moles of solute in the solution does not change, but the concentration of the solution decreases and the volume of the solution increases.

19. B
Let the concentration of the diluted solution be M_2.
$2.0 \times \frac{50.0}{1000} = M_2 \times \frac{250.0}{1000}$
$M_2 = 0.4$
∴ the concentration of the diluted solution is 0.4 mol dm^{-3}.

20. B
Let the volume of the diluted solution be V_2.
$2.0 \times \frac{100}{1000} = 0.5 \times V_2$
$V_2 = 0.4$
Volume of diluted solution obtained = 0.4 dm^3 (or 400 cm^3)
∴ the volume of water needed = (400 – 100) cm^3 = 300 cm^3

21. B
Let the volume of 0.50 M $H_2SO_4(aq)$ needed be V_1.
$0.50 \times V_1 = 0.15 \times \frac{250.0}{1000}$
$V_1 = 0.075$
∴ the volume of 0.50 M $H_2SO_4(aq)$ needed is 0.075 dm^3 or 75 cm^3.

D. Structured questions (p.67)

22. (a) Concentration of $K_2CO_3(aq)$ in mol dm^{-3}
$= \frac{0.350 \text{ mol}}{\frac{250.0}{1000} \text{ dm}^3} = 1.40 \text{ mol dm}^{-3}$ 1
(b) Molar mass of K_2CO_3
$= (39.1 \times 2 + 12.0 + 16.0 \times 3) \text{ g mol}^{-1}$
$= 138.2 \text{ g mol}^{-1}$
Concentration of $K_2CO_3(aq)$ in g dm^{-3}
$= 1.40 \text{ mol dm}^{-3} \times 138.2 \text{ g mol}^{-1}$
$= 193 \text{ g dm}^{-3}$ 1

23. (a) Number of moles of $Mg(NO_3)_2$
$= 2.0 \times \frac{125.0}{1000}$ mol = 0.25 mol 1
(b) Molar mass of $Mg(NO_3)_2$
$= [24.3 + (14.0 + 16.0 \times 3) \times 2] \text{ g mol}^{-1}$
$= 148.3 \text{ g mol}^{-1}$
Mass of $Mg(NO_3)_2$
$= 0.25 \text{ mol} \times 148.3 \text{ g mol}^{-1} = 37 \text{ g}$ 1

24. Volume of the resultant solution
$= (100.0 + 200.0 + 100.0) \text{ cm}^3$
$= 400.0 \text{ cm}^3 = 0.4 \text{ dm}^3$
(a) Molarity of $Na^+(aq)$ in the resultant solution
$= \frac{(1.5 \times \frac{100.0}{1000} \times 2 + 2.0 \times \frac{200.0}{1000}) \text{ mol}}{0.4 \text{ dm}^3}$
$= 1.75 \text{ mol dm}^{-3}$ 1
(b) Molarity of $K^+(aq)$ in the resultant solution
$= \frac{1.0 \times \frac{100.0}{1000} \text{ mol}}{0.4 \text{ dm}^3} = 0.25 \text{ mol dm}^{-3}$ 1
(c) Molarity of $Cl^-(aq)$ in the resultant solution
$= \frac{(2.0 \times \frac{200.0}{1000} + 1.0 \times \frac{100.0}{1000}) \text{ mol}}{0.4 \text{ dm}^3}$
$= 1.25 \text{ mol dm}^{-3}$ 1

Chapter 16

A. Fill in the blanks (p.86)

1. (a) higher
(b) equal
(c) lower
2. pH; $-\log[H^+(aq)]$
3. lower; alkalinity
4. pH; methyl orange; litmus; phenolphthalein
5. Universal; Universal; pH
6. pH meter; pH sensor

B. Practice questions (p.86)

7. (a) pH = –log 0.002 = 2.7; the solution is acidic.
(b) pH = $-\log 1.8 \times 10^{-5}$ = 4.7; the solution is acidic.
(c) pH = $-\log 1.0 \times 10^{-7}$ = 7.0; the solution is neutral.
(d) pH = $-\log 7.2 \times 10^{-9}$ = 8.1; the solution is alkaline.

8. (a) $5.0 = -\log[H^+(aq)]$
$[H^+(aq)] = 10^{-5.0} \text{ M} = 1.0 \times 10^{-5} \text{ M}$
(b) $12.5 = -\log[H^+(aq)]$
$[H^+(aq)] = 10^{-12.5} \text{ M} = 3.2 \times 10^{-13} \text{ M}$
(c) $5.6 = -\log[H^+(aq)]$
$[H^+(aq)] = 10^{-5.6} \text{ M} = 2.5 \times 10^{-6} \text{ M}$
(d) $9.0 = -\log[H^+(aq)]$
$[H^+(aq)] = 10^{-9.0} \text{ M} = 1.0 \times 10^{-9} \text{ M}$

9.

Indicator	Colour of the indicator in		
	acidic solution (pH 3)	neutral solution (pH 7)	alkaline solution (pH 9)
Methyl orange	red	yellow	yellow
Litmus	red	purple	blue
Phenolphthalein	colourless	colourless	very pale pink

C. Multiple-choice questions (p.87)

10. B
(1) and (3): for both 0.10 M HCl(aq) and 0.10 M HNO_3(aq), their pH = –log 0.10 = 1.0.
(2): assume that H_2SO_4 ionizes completely in water according to the following equation:
$H_2SO_4(aq) \longrightarrow 2H^+(aq) + SO_4^{2-}(aq)$
From the equation, 1 mol of H_2SO_4 ionizes to give 2 mol of H^+(aq).
∴ pH of 0.10 M H_2SO_4(aq) = –log (0.10 × 2) = 0.70

11. C
(1) and (2): the pH of 0.01 M HCl(aq) and 0.01 M H_2SO_4(aq) are 2 and 1.7 respectively.
(3): NH_3(aq) is alkaline. Its pH is greater than 7.
(4): H_2O(l) is neutral, its pH is equal to 7.

12. A
When an acid is diluted by adding water, the concentration of H^+(aq) ions in the acid decreases and hence the pH of the acid increases. However, the number of moles of H^+(aq) ions in the acid remains unchanged.

13. A

14. C
The pH of 0.1 M NaOH(aq) is 13.

15. B

16. D
The pH of glass cleaner is around 11. Refer to p.83 for the detail.

17. B

18. A
Refer to p.83 for the pH of some common substances.

19. C
Calcium hydroxide solution is alkaline. Its pH is greater than 7.

20. A
The pH of lemon juice is around 2.5.

D. Structured questions (p.88)

21. (a) Any TWO of the following methods:
Use pH paper/universal indicator/pH meter/data-logger connected with a pH sensor. 2
(b) Oven cleaner, distilled water, orange juice 1
(c) Orange juice 1

Chapter 17

A. Fill in the blanks (p.104)

1. (a) completely; almost completely; hydrochloric acid/nitric acid/sulphuric acid
(b) slightly; ethanoic acid/carbonic acid/sulphurous acid
2. (a) dissociates; sodium hydroxide/potassium hydroxide
(b) slightly; aqueous ammonia
3. (a) higher
(b) lower
(c) higher
4. (a) dilute; strong
(b) concentrated; weak
(c) concentrated; strong
(d) dilute; weak

B. Multiple-choice questions (p.104)

5. D

6. C
Option (C): as shown in the diagram, the molecules of the acid ionize completely in water to give H^+(aq) ions and the corresponding anions.

7. D

8. B

9. B

C. Structured questions (p.105)

10. (a) pH = –log 0.01 = 2 1
(b) Higher than that of 0.01 M hydrochloric acid
Ethanoic acid is a weak acid. It ionizes only slightly in water. 1

As $[H^+(aq)]$ in 0.01 M ethanoic acid is smaller than 0.01 M, the pH of this acid is greater than 2. 1

(c) 0.01 M hydrochloric acid has a higher electrical conductivity.
Hydrochloric acid is a strong acid while ethanoic acid is a weak acid. The concentration of mobile ions in 0.01 M hydrochloric acid is higher compared with that in 0.01 M ethanoic acid. 1
The higher the concentration of mobile ions present in an acid, the higher is the electrical conductivity of the acid. 1

(d) 0.01 M hydrochloric acid reacts more vigorously with the iron nail.
Hydrochloric acid is a strong acid while ethanoic acid is a weak acid. The concentration of $H^+(aq)$ ions in 0.01 M hydrochloric acid is higher compared with 0.01 M ethanoic acid. 1
An acid with a higher concentration of $H^+(aq)$ ions would react more vigorously with an iron nail. 1

11. (a) $[H^+(aq)] = 10^{-3.0}$ M = 0.001 M 1

(b) Ethanoic acid is a weak acid, it ionizes slightly in water to form $H^+(aq)$ ions. Hence, if the concentration of $H^+(aq)$ ions in a sample of ethanoic acid is 0.001 M, the concentration of ethanoic acid in the sample must be larger than 0.001 M. 1

Chapter 18

A. Fill in the blanks (p.142)

1. base
2. exothermic
3. (a) pH
   (b) acid
   (c) waste
   (d) fertilizers
4. ionizable; metal; ammonium
5. (a) metal; base; carbonate
   (b) alkali; soluble
6. precipitation

B. Practice questions (p.142)

7. (a) (i) $2HCl(aq) + Ca(OH)_2(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$
   (ii) $H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$
   (iii) Calcium chloride
   (b) (i) $2HNO_3(aq) + CaO(s) \longrightarrow Ca(NO_3)_2(aq) + H_2O(l)$
   (ii) $2H^+(aq) + CaO(s) \longrightarrow Ca^{2+}(aq) + H_2O(l)$
   (iii) Calcium nitrate
   (c) (i) $H_2SO_4(aq) + Mg(OH)_2(s) \longrightarrow MgSO_4(aq) + 2H_2O(l)$
   (ii) $2H^+(aq) + Mg(OH)_2(s) \longrightarrow Mg^{2+}(aq) + 2H_2O(l)$
   (iii) Magnesium sulphate
   (d) (i) $HNO_3(aq) + NH_3(aq) \longrightarrow NH_4NO_3(aq)$
   (ii) $H^+(aq) + NH_3(aq) \longrightarrow NH_4^+(aq)$
   (iii) Ammonium nitrate

8.

Salt	Parent acid	Base
(a) Ammonium chloride	HCl	NH_3
(b) Copper(II) nitrate	HNO_3	$CuO/Cu(OH)_2$ (any one)
(c) Iron(II) sulphate	H_2SO_4	$FeO/Fe(OH)_2$ (any one)
(d) Potassium ethanoate	CH_3COOH	K_2O/KOH (any one)

9.

	K^+	Na^+	Ca^{2+}	Al^{3+}	Zn^{2+}	Fe^{2+}	Pb^{2+}	Cu^{2+}	NH_4^+	Ag^+	Ba^{2+}
NO_3^-											
HCO_3^-											
CO_3^{2-}			$CaCO_3$	Al_2CO_3	$ZnCO_3$	$FeCO_3$	$PbCO_3$	$CuCO_3$		Ag_2CO_3	$BaCO_3$
Cl^-							$PbCl_2$			AgCl	
Br^-							$PbBr_2$			AgBr	
I^-							PbI_2			AgI	
SO_4^{2-}			$CaSO_4$				$PbSO_4$				$BaSO_4$

10. (a) By reacting excess magnesium metal/ magnesium oxide/magnesium carbonate/ magnesium hydroxide with dilute sulphuric acid

$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$/
$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$

OR

$MgO(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2O(l)$/
$MgO(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2O(l)$

OR

$MgCO_3(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2O(l) + CO_2(g)$/
$MgCO_3(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2O(l) + CO_2(g)$

OR

$Mg(OH)_2(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + 2H_2O(l)$/
$Mg(OH)_2(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + 2H_2O(l)$

(b) By precipitation between lead(II) nitrate solution and sodium sulphate solution

$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow PbSO_4(s) + 2NaNO_3(aq)$/
$Pb^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s)$

(c) By titrating potassium hydroxide solution/ potassium carbonate solution with dilute nitric acid

$KOH(aq) + HNO_3(aq) \longrightarrow KNO_3(aq) + H_2O(l)$/
$OH^-(aq) + H^+(aq) \longrightarrow H_2O(l)$

OR

$K_2CO_3(aq) + 2HNO_3(aq) \longrightarrow 2KNO_3(aq) + H_2O(l) + CO_2(g)$/
$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$

C. Multiple-choice questions (p.143)

11. C
12. D
13. A

The volume of the reaction mixture is the same in both experiments, but the number of moles of $H^+(aq)$ ions and $OH^-(aq)$ ions reacting in Experiment 2 is twice that in Experiment 1. Twice as much heat is given out in Experiment 2 compared with that in Experiment 1. Therefore, the maximum temperature rise of neutralization reaction in Experiment 2 is greater than that in Experiment 1.

14. B

Calcium sulphate does not neutralize the acid in soil as it is not an alkali or alkaline substance.

15. C

Although aqueous ammonia can be used to neutralize the acidic substances in liquid waste, it is seldom used to treat acidic liquid waste from industries. This is because aqueous ammonia is relative expensive and not easy to handle. Vapour of aqueous ammonia is extremely corrosive to nose, throat and mucous membrane.

16. B
17. D
18. C
19. C
20. D

Zinc metal, zinc oxide, zinc hydroxide and zinc carbonate are all insoluble in water. No aqueous solutions of these substances can be prepared for titration.

21. D
22. B

For (2): the equation for the reaction involved is $NH_3(aq) + HNO_3(aq) \longrightarrow NH_4NO_3(aq)$.

23. C

Zinc carbonate ($ZnCO_3$) is insoluble in water. So we do not write the ions separately.

24. D

Lead(II) chloride is insoluble in water. It can be prepared by precipitation:
$Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow PbCl_2(s)$
(The $Pb^{2+}(aq)$ ions come from lead(II) nitrate solution.)

25. A

Option (B): $Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$
Option (C): $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow PbCl_2(s)$
Option (D): $Ca^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow CaSO_4(s)$

D. Structured questions (p.145)

26. (a) Calcium hydroxide 1
(b) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$ 1
$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(s)$ 1
(c) (i) It can neutralize the acid in soil. 1
(ii) It can neutralize the acidic substances in the industrial liquid waste. 1
(d) $Ca(OH)_2(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + 2H_2O(l)$ 1

27. (a) Silver is not a reactive metal. It does not react with dilute hydrochloric acid. 1
(b) Silver chloride can be prepared by precipitation between silver nitrate solution and sodium chloride solution. 1
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$ 1

28. (a) By reacting excess $Cu(OH)_2(s)$ with $H_2SO_4(aq)$ 1

$Cu(OH)_2(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + 2H_2O(l)$ 1

(b) By precipitation between $CaCl_2(aq)$ and $K_2SO_4(aq)$ (or $H_2SO_4(aq)$) 1

$CaCl_2(aq) + K_2SO_4(aq) \longrightarrow CaSO_4(s) + 2KCl(aq)$ 1

OR

$CaCl_2(aq) + H_2SO_4(aq) \longrightarrow CaSO_4(s) + 2HCl(aq)$ (1)

(c) By titrating $Na_2CO_3(aq)$ with $H_2SO_4(aq)$ 1

$Na_2CO_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l)$ 1

Chapter 19

A. Fill in the blanks (p.192)

1. primary standard
2. standard solution; primary standard; known
3. pipettes; burettes; distilled water; Volumetric flasks; conical flasks; distilled water
4. Equivalence point
5. end; strength
   (a) methyl orange; phenolphthalein
   (b) methyl orange
   (c) phenolphthalein
6. equivalence

B. Practice questions (p.192)

7. Number of moles of $Na_2CO_3 \cdot 10H_2O$

$$= \frac{14.3}{(23.0 \times 2 + 12.0 + 16.0 \times 3) + (1.0 \times 2 + 16.0) \times 10} \text{ mol}$$

$= 0.0500$ mol

1 mol of $Na_2CO_3 \cdot 10H_2O$ contains 1 mol of Na_2CO_3.

$\therefore$ number of moles of Na_2CO_3 in the 250.0 cm^3 solution $= 0.0500$ mol

Molarity of $Na_2CO_3(aq) = \dfrac{0.0500}{\frac{250.0}{1000}}$ mol dm^{-3}

$= 0.200$ mol dm^{-3}

8. Let the volume of 2.0 M HCl(aq) required be V_1.

$2.0 \times V_1 = 0.15 \times \dfrac{250.0}{1000}$

$V_1 = 0.019$

$\therefore$ the volume of 2.0 M HCl(aq) required is 0.019 dm^3 or 19 cm^3.

9. $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

Number of moles of NaOH $= 0.10 \times \dfrac{25.0}{1000}$ mol

$= 2.5 \times 10^{-3}$ mol

From the equation, mole ratio of NaOH to HCl = 1 : 1.

$\therefore$ molarity of HCl(aq) $= \dfrac{2.5 \times 10^{-3}}{\frac{22.0}{1000}}$ mol dm^{-3}

$= 0.11$ mol dm^{-3}

10. Number of moles of $(COOH)_2 \cdot 2H_2O$ crystals used

$$= \frac{3.78}{(12.0 + 16.0 \times 2 + 1.0) \times 2 + (1.0 \times 2 + 16.0) \times 2} \text{ mol}$$

$= 0.0300$ mol

1 mol of $(COOH)_2 \cdot 2H_2O$ contains 1 mol of $(COOH)_2$.

$\therefore$ number of moles of $(COOH)_2$ in 25.0 cm^3 of the diluted solution

$= 0.0300 \times \dfrac{25.0}{250.0}$ mol $= 0.00300$ mol

$(COOH)_2(aq) + 2KOH(aq) \longrightarrow (COO)_2K_2(aq) + 2H_2O(l)$

From the equation, mole ratio of $(COOH)_2$ to KOH = 1 : 2.

$\therefore$ molarity of KOH(aq) $= \dfrac{0.00300 \times 2}{\frac{15.00}{1000}}$ mol dm^{-3}

$= 0.400$ mol dm^{-3}

11. Number of moles of the acid in the 25.0 cm^3 acid solution

$= \dfrac{1.89}{126.0} \times \dfrac{25.0}{250.0}$ mol $= 1.50 \times 10^{-3}$ mol

Number of moles of NaOH used

$= 0.200 \times \dfrac{15.0}{1000}$ mol $= 3.00 \times 10^{-3}$ mol

$\dfrac{\text{Number of moles of NaOH}}{\text{Number of moles of the acid}} = \dfrac{3.00 \times 10^{-3}}{1.50 \times 10^{-3}} = 2$

$\therefore$ the basicity of the acid is 2.

12. $2MHCO_3(aq) + H_2SO_4(aq) \longrightarrow M_2SO_4(aq) + 2CO_2(g) + 2H_2O(l)$

Number of moles of H_2SO_4 used

$= 0.20 \times \dfrac{10.50}{1000}$ mol $= 2.1 \times 10^{-3}$ mol

From the equation, mole ratio of H_2SO_4 to $MHCO_3$ = 1 : 2.

$\therefore$ number of moles of $MHCO_3$ in the 100.0 cm^3 solution

$= 2.1 \times 10^{-3} \times 2 \times \dfrac{100.0}{25.0}$ mol $= 0.0168$ mol

Let the relative atomic mass of M be x.

$0.0168 = \dfrac{1.682}{x + 1.0 + 12.0 + 16.0 \times 3}$

$x = 39.1$

$\therefore$ the relative atomic mass of M is 39.1.

13. $H_3A(aq) + 3NaOH(aq) \longrightarrow Na_3A(aq) + 3H_2O(l)$

Number of moles of NaOH used

$= 0.200 \times \dfrac{18.75}{1000}$ mol $= 3.75 \times 10^{-3}$ mol

From the equation, mole ratio of NaOH to H_3A = 3 : 1.

∴ number of moles of H_3A in the 500.0 cm^3 of solution

$= \frac{3.75 \times 10^{-3}}{3} \times \frac{500.0}{25.0}$ mol = 0.025 mol

1 mol of $H_3A \cdot nH_2O$ contains 1 mol of H_3A. Hence, number of moles of $H_3A \cdot nH_2O$ in 5.25 g $H_3A \cdot nH_2O$ = 0.025 mol

$0.025 = \frac{5.25}{(192.0 + 18n)}$

$n = 1$

14. $H_2SO_4(aq) + Na_2CO_3(aq) \longrightarrow Na_2SO_4(aq) + CO_2(g) + H_2O(l)$

Number of moles of H_2SO_4 used

$= 0.5 \times \frac{20.00}{1000}$ mol = 0.01 mol

From the equation, mole ratio of H_2SO_4 to Na_2CO_3 = 1 : 1.

∴ number of moles of $Na_2CO_3 \cdot 10H_2O$ in the mixture = 0.01 mol

Percentage by mass of $Na_2CO_3 \cdot 10H_2O$ in the mixture =

$\frac{0.01 \times [(23.0 \times 2 + 12.0 + 16.0 \times 3) + (1.0 \times 2 + 16.0) \times 10]}{3.06} \times 100\%$

= 93.5%

15. $2HNO_3(aq) + CaCO_3(s) \longrightarrow Ca(NO_3)_2(aq) + CO_2(g) + H_2O(l)$

Number of moles of HNO_3 reacted with $CaCO_3$ in the sample

$= (0.50 \times \frac{50.0}{1000} - 0.40 \times \frac{20.0}{1000})$ mol = 0.017 mol

From the equation, mole ratio of HNO_3 to $CaCO_3$ = 2 : 1.

∴ number of moles of $CaCO_3$ in the sample

$= \frac{0.017}{2}$ mol $= 8.5 \times 10^{-3}$ mol

Percentage by mass of $CaCO_3$ in the sample

$= \frac{8.5 \times 10^{-3} \times (40.1 + 12.0 + 16.0 \times 3)\ g}{1.28\ g} \times 100\%$

= 66.5%

C. Multiple-choice questions (p.193)

16. C

17. B

Let the molarity of the diluted $Na_2CO_3(aq)$ be M_2.

$0.80 \times \frac{10.0}{1000} = M_2 \times \frac{250.0}{1000}$

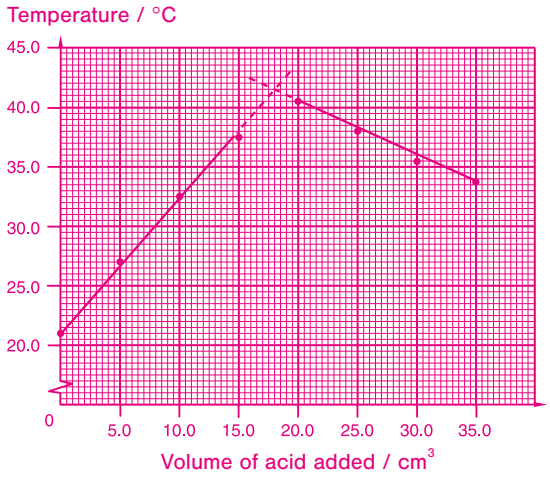
$M_2 = 0.032$

∴ the molarity of the diluted $Na_2CO_3(aq)$ is 0.032 M.

18. B

Burettes are used to deliver variable volumes of a solution accurately.

19. C



20. B

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

From the equation, mole ratio of NaOH to HCl = 1 : 1.

∴ molarity of HCl(aq) $= \frac{0.1 \times \frac{25.0}{1000}}{\frac{18.00}{1000}}$ M = 0.139 M

21. A

$H_3A(aq) + 3KOH(aq) \longrightarrow K_3A(aq) + 3H_2O(l)$

From the equation, mole ratio of KOH to H_3A = 3 : 1.

∴ concentration of the acid

$= \frac{0.5 \times \frac{21.35}{1000} \times \frac{1}{3}}{\frac{25.0}{1000}}$ M = 0.14 M

22. B

$\frac{\text{Number of moles of NaOH}}{\text{Number of moles of the acid}} = \frac{2.5 \times \frac{40.00}{1000}}{2.0 \times \frac{25.0}{1000}}$

$= \frac{0.10}{0.050} = 2$

23. B

$H_3A(aq) + 3NaOH(aq) \longrightarrow Na_3A(aq) + 3H_2O(l)$

From the equation, mole ratio of NaOH to H_3A = 3 : 1.

∴ molar mass of H_3A

$= \frac{0.108}{0.12 \times \frac{27.60}{1000} \times \frac{1}{3}}$ g mol^{-1} = 97.8 g mol^{-1}

Answers to Exam practice

Chapter 14

A. Multiple-choice questions (p.45)

1. B

$Cu(OH)_2(s) + 2HCl(aq) \longrightarrow CuCl_2(aq) + 2H_2O(l)$

2. D
$MgCO_3(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + CO_2(g) + H_2O(l)$
$MgCl_2(aq)$ is colourless.

3. C
$Ca(OH)_2(aq) + 2HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$
Neutralization reaction is exothermic, so the solution becomes warm.

4. D (80%)

5. D
$H_2C_2O_4(aq) \rightleftharpoons 2H^+(aq) + C_2O_4^{2-}(aq)$

6. D
(2): $NH_4Cl(aq) + KOH(aq) \longrightarrow KCl(aq) + NH_3(g) + H_2O(l)$
(3): lemon juice contains $H^+(aq)$; $HCO_3^-(aq) + H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$

7. B (65%)
When a small amount of NaOH(aq) is added to $Ca^{2+}(aq)$, a white precipitate forms, which is insoluble in excess NaOH(aq).

8. A
Statement (2): silver oxide dissolves in excess $NH_3(aq)$, but not in excess NaOH(aq).
Statement (3): both NaOH(aq) and $NH_3(aq)$ are alkalis. Both can react with carbon dioxide which is an acidic oxide.

9. B
The following metal hydroxides are insoluble in excess sodium hydroxide solution: $Mg(OH)_2$, $Ca(OH)_2$
The following metal hydroxides are soluble in excess sodium hydroxide solution: $Zn(OH)_2$, $Pb(OH)_2$, $Al(OH)_3$

10. C
When a small amount of sodium hydroxide solution is added to zinc carbonate solution, the following reaction occurs:
$ZnCO_3(s) + 2NaOH(aq) \longrightarrow Zn(OH)_2(s) + Na_2CO_3(aq)$

11. A
Dilute hydrochloric acid conducts electricity because it contains mobile ions, which are produced by the ionization of hydrogen chloride when dissolved in water.

B. Structured questions (p.46)

12. (a) 1 1
(b) $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$ 1
(c) (i) ① Magnesium dissolves. Colourless gas bubbles form. 1
② Sodium carbonate powder dissolves. Colourless gas bubbles form/ effervescence occurs. 1
③ No observable change 1
(ii) (1) $Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$ 1
(2) $CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow CO_2(g) + H_2O(l)$ 1
(iii) Ethanoic acid ionizes (slightly) in water to produce mobile ions. Hence, Solution *X* conducts electricity. 1
Tetrachloromethane does not contain water. Ethanoic acid does not ionize in the absence of water. As there are no mobile ions in Solution *Y*, it does not conduct electricity. 1

13. (a) $(NH_4)_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2NH_3(g) + 2H_2O(l)$ 1
OR
$NH_4^+(aq) + OH^-(aq) \longrightarrow NH_3(g) + H_2O(l)$ (1)
(b) Set-up 1
This is because ammonia is less dense than air and is very soluble in water. 1
(c) Ammonia gas is toxic if inhaled. 1
OR
Exposure to ammonia gas may cause (severe) skin burns and eye damage. (1)

14. (a) Add solid calcium hydroxide to water and stir the mixture for a period of time. 1
Then filter the suspension to get the clear filtrate. 1
(b) Hydroxide ion/$OH^-(aq)$ 1
(c) $2NH_4NO_3(aq) + Ca(OH)_2(aq) \longrightarrow Ca(NO_3)_2(aq) + 2NH_3(g) + 2H_2O(l)$ 1
OR
$NH_4^+(aq) + OH^-(aq) \longrightarrow NH_3(g) + H_2O(l)$ (1)
(d) A pale blue precipitate forms. 1
$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$ 1

15. Add aqueous ammonia to each of the solutions until in excess. 1
A white precipitate forms initially in each solution. But only the precipitate formed by $Zn^{2+}(aq)$ dissolves in excess aqueous ammonia. 1
Add sodium hydroxide solution to each of the remaining solutions until in excess. 1
A white precipitate forms initially in each solution. But only the precipitate formed by $Pb^{2+}(aq)$ dissolves in excess sodium hydroxide solution. 1

Communication mark 1

(Note: the white precipitate formed by $Mg^{2+}(aq)$ neither dissolves in excess aqueous ammonia nor sodium hydroxide solution.)

16. Dissolve a sample of the three solids separately in water. 1

Add sodium hydroxide solution to each of the solutions obtained until in excess. 1

A white precipitate forms initially in each solution. But only the precipitate of $ZnCl_2$ dissolves in excess sodium hydroxide solution. 1

Heat respectively a sample of the remaining two solids in a test tube and place a piece of dry cobalt(II) chloride paper near the mouth of the test tube. 1

Only $CaCl_2 \cdot 6H_2O$ can turn the paper from blue to pink. 1

Communication mark 1

Chapter 15

A. Multiple-choice questions (p.68)

1. D

Number of moles of $C_8H_5O_4K$

$= \dfrac{1.63}{12.0 \times 8 + 1.0 \times 5 + 16.0 \times 4 + 39.1}$ mol

$= 7.99 \times 10^{-3}$ mol

Concentration of $C_8H_5O_4K(aq)$

$= \dfrac{7.99 \times 10^{-3}}{\frac{100.0}{1000}}$ mol dm^{-3} = 0.0799 mol dm^{-3}

2. C

Number of moles of oxalic acid crystals

$= \dfrac{4.81}{[(12.0 + 16.0 \times 2 + 1.0) \times 2 + (1.0 \times 2 + 16.0) \times 2]}$ mol

= 0.0382 mol

Molarity of the oxalic acid solution

$= \dfrac{0.0382}{\frac{250.0}{1000}}$ mol dm^{-3} = 0.153 mol dm^{-3}

3. D

Number of moles of NH_3

$= \dfrac{150}{14.0 + 1.0 \times 3}$ mol = 8.82 mol

Concentration of $NH_3(aq)$

$= \dfrac{8.82}{\frac{500}{1000}}$ mol dm^{-3} = 17.6 mol dm^{-3}

(The author is responsible for the solutions and that (a) they have neither been provided nor approved by AQA and (b) they may not necessarily constitute the only possible solutions.)

4. D

Number of moles of $Fe_2(SO_4)_3$

$= 0.200 \times \dfrac{250.0}{1000}$ mol = 0.0500 mol

Mass of $Fe_2(SO_4)_3$

= 0.0500 mol $\times$ [55.8 $\times$ 2 + (32.1 + 16.0 $\times$ 4) $\times$ 3] g mol^{-1}

= 20.0 g

5. A

Option (A):

Number of moles of Na_2SO_4

$= \dfrac{5.0}{23.0 \times 2 + 32.1 + 16.0 \times 4}$ mol = 0.0352 mol

Number of moles of SO_4^{2-} = 0.0352 mol

Concentration of $SO_4^{2-}(aq)$

$= \dfrac{0.0352}{\frac{250.0}{1000}}$ mol dm^{-3} = 0.141 mol dm^{-3}

Option (B):

Number of moles of K_2SO_4

$= \dfrac{5.0}{39.1 \times 2 + 32.1 + 16.0 \times 4}$ mol = 0.0287 mol

Number of moles of SO_4^{2-} = 0.0287 mol

Concentration of $SO_4^{2-}(aq)$

$= \dfrac{0.0287}{\frac{250.0}{1000}}$ mol dm^{-3} = 0.115 mol dm^{-3}

Option (C):

Number of moles of $CuSO_4$

$= \dfrac{5.0}{63.5 + 32.1 + 16.0 \times 4}$ mol = 0.0313 mol

Number of moles of SO_4^{2-} = 0.0313 mol

Concentration of $SO_4^{2-}(aq)$

$= \dfrac{0.0313}{\frac{250.0}{1000}}$ mol dm^{-3} = 0.125 mol dm^{-3}

Option (D):

Number of moles of $FeSO_4$

$= \dfrac{5.0}{55.8 + 32.1 + 16.0 \times 4}$ mol = 0.0329 mol

Number of moles of SO_4^{2-} = 0.0329 mol

Concentration of $SO_4^{2-}(aq)$

$= \dfrac{0.0329}{\frac{250.0}{1000}}$ mol dm^{-3} = 0.132 mol dm^{-3}

6. B

Solution (1):

There are 1 Na^+ ion and 1 NO_3^- ion in each formula unit of $NaNO_3$.

Total number of moles of ions

$= 1.0 \times \dfrac{36}{1000} \times 2$ mol = 0.072 mol

Solution (2):

There are 1 Na^+ ion and 1 OH^- ion in each formula unit of NaOH.

Total number of moles of ions

$= 1.5 \times \frac{30}{1000} \times 2 \text{ mol} = 0.090 \text{ mol}$

Solution (3):

There are 2 K^+ ions and 1 SO_4^{2-} ion in each formula unit of K_2SO_4.

Total number of moles of ions

$= 1.0 \times \frac{24}{1000} \times 3 \text{ mol} = 0.072 \text{ mol}$

7. C

Number of moles of SO_4^{2-} from K_2SO_4 = 0.05 mol

Number of moles of SO_4^{2-} from $Al_2(SO_4)_3$

= 0.05 mol × 3 = 0.15 mol

Total number of moles of SO_4^{2-}

= (0.05 + 0.15) mol = 0.20 mol

Molarity of $SO_4^{2-}(aq)$ ions

$= \frac{0.20}{\frac{250.0}{1000}} \text{ mol dm}^{-3} = 0.80 \text{ mol dm}^{-3}$

8. A

There are 3 Na^+ ions in each formula unit of Na_3PO_4.

Hence, the number of moles of Na_3PO_4 that contains 0.03 mol of Na^+ ions

$= \frac{0.03}{3} \text{ mol} = 0.01 \text{ mol}$

Molarity of $Na_3PO_4(aq)$

$= \frac{0.01}{\frac{50.0}{1000}} \text{ mol dm}^{-3} = 0.2 \text{ mol dm}^{-3}$

9. B

Let the molarity of the remaining solution be M_2.

$0.05 \times \frac{250.0}{1000} = M_2 \times \frac{220.0}{1000}$

$M_2 = 0.057$

∴ the molarity of the remaining solution was 0.057 M.

10. B

Let the volume of 10.0 M HCl(aq) needed be V_1.

$10.0 \times V_1 = 0.30 \times \frac{500.0}{1000}$

$V_1 = 0.015 \text{ dm}^3$

∴ the volume of 10.0 M HCl(aq) needed is 0.015 dm^3 (or 15 cm^3).

11. C

Number of moles of $MgSO_4$

$= \frac{1.204}{24.3 + 32.1 + 16.0 \times 4} \text{ mol} = 0.01000 \text{ mol}$

Volume of the diluted $MgSO_4(aq)$

$= \frac{0.01000 \text{ mol}}{0.02 \text{ mol dm}^{-3}} = 0.5 \text{ dm}^3 \text{ (or } 500 \text{ cm}^3\text{)}$

B. Structured questions (p.69)

12. (a) Number of moles of sucrose

$= \frac{5.00 \text{ g}}{(12.0 \times 12 + 1.0 \times 22 + 16.0 \times 11) \text{ g mol}^{-1}}$

$= 0.0146 \text{ mol}$ 1

Concentration limit of sugar in 'low sugar' drinks

$= \frac{0.0146 \text{ mol}}{\frac{100}{1000} \text{ dm}^3} = 0.146 \text{ mol dm}^{-3}$ 1

(b) Mass of sugar per 100 ml of that carton of lemon tea

$= \frac{26.5 \text{ g}}{375 \text{ ml}} \times 100 \text{ ml} = 7.07 \text{ g}$ 1

As that carton of lemon tea contains more than 5.00 g of sugar per 100 ml of the lemon tea, it cannot claim to be 'low sugar'. 1

13. (a) Mass of NH_3 present in 1 dm^3 of this solution = 0.88 g cm^{-3} × 1000 cm^3 × 35%

= 308 g 1

(b) Molarity of this solution

$= \frac{308 \text{ g}}{(14.0 + 1.0 \times 3) \text{ g mol}^{-1}} = 18.1 \text{ mol dm}^{-3}$ 1

14. (a) $Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$ 1

(b) Some shiny silvery solid deposits on the copper surface. 1

The solution changes from colourless to blue. 1

(c) Number of moles of Cu used

$= \frac{0.635}{63.5} \text{ mol} = 0.0100 \text{ mol}$ 1

Number of moles of Ag^+ ions in 100 cm^3 of 0.5 M $AgNO_3$

$= 0.5 \times \frac{100}{1000} \text{ mol} = 0.05 \text{ mol}$ 1

0.0100 mol of Cu requires only 0.0100 mol × 2 = 0.0200 mol of Ag^+ for complete reaction. Hence, Cu is the limiting reactant. In the resultant solution,

concentration of $Cu^{2+}(aq)$ ions

$= \frac{0.01 \text{ mol}}{\frac{100}{1000} \text{ dm}^3} = 0.1 \text{ mol dm}^{-3}$ 1

concentration of $Ag^+(aq)$ ions

$= \frac{(0.05 - 0.02) \text{ mol}}{\frac{100}{1000} \text{ dm}^3} = 0.3 \text{ mol dm}^{-3}$ 1

15. Number of moles of OH^- in the 400 cm^3 solution
$= 6.56 \times 10^{-3}$ mol $dm^{-3} \times \frac{400}{1000}$ dm^3
$= 2.624 \times 10^{-3}$ mol 1
Number of moles of $M(OH)_2$ in the 400 cm^3 solution
$= \frac{2.624 \times 10^{-3}}{2}$ mol $= 1.312 \times 10^{-3}$ mol 1
From the equation, mole ratio of $M(OH)_2$ to M = 1 : 1.
∴ molar mass of M
$= \frac{0.115 \text{ g}}{1.312 \times 10^{-3} \text{ mol}} = 87.7 \text{ g mol}^{-1}$
M is strontium. 1
(Note: from the Periodic Table, the relative atomic mass of strontium is 87.6.)

16. (a) Mass of NaOCl in the 10 cm^3 of chlorine bleach sample
$= 10 \text{ cm}^3 \times 1.0 \text{ g cm}^{-3} \times 6.0\% = 0.60 \text{ g}$
Number of moles of NaOCl in the 10 cm^3 of chlorine bleach sample
$= \frac{0.60 \text{ g}}{(23.0 + 16.0 + 35.5) \text{ g mol}^{-1}}$
$= 8.05 \times 10^{-3}$ mol 1
Molarity of NaOCl(aq) in the original sample
$= \frac{8.05 \times 10^{-3} \text{ mol}}{\frac{10}{1000} \text{ dm}^3} = 0.805 \text{ mol dm}^{-3}$ 1

(b) Approximate molarity of the 1:99 disinfectant solution
$= \frac{8.05 \times 10^{-3} \text{ mol}}{\frac{10 + 990}{1000} \text{ dm}^3}$ 1
$= 8.05 \times 10^{-3}$ mol dm^{-3} 1

Chapter 16

A. Multiple-choice questions (p.89)

1. B
Both *P* and *Q* are acids, their pH are less than 7. HCl(aq) is a monobasic acid, the $[H^+(aq)]$ in 0.01 M HCl(aq) is 0.01 M. $H_2SO_4(aq)$ is a dibasic acid, the $[H^+(aq)]$ in 0.01 M $H_2SO_4(aq)$ is higher than 0.01 M. Hence, 0.01 M $H_2SO_4(aq)$ is more acidic than 0.01 M HCl(aq), and the pH of *Q* is lower than that of *P*. Both *R* and *S* are alkalis, their pH are greater than 7. As 0.01 M NaOH(aq) has a lower $[OH^-(aq)]$ than 0.10 M NaOH(aq), 0.01 M NaOH(aq) is less alkaline than 0.10 M NaOH(aq), and hence the pH of *R* is lower than that of *S*.

2. B
As shown in Table 14.2 on p.17, nitric acid ionizes completely in water but ethanoic acid does not. For the same molar concentration, nitric acid has a higher $[H^+(aq)]$ than ethanoic acid. Hence, 0.10 mol dm^{-3} nitric acid has a lower pH.

3. B
1 mol of H_2SO_4 ionizes almost completely in water to give 2 mol of $H^+(aq)$ and 1 mol of $SO_4^{2-}(aq)$.
∴ $[H^+(aq)] = 0.03 \times 2$ mol $dm^{-3} = 0.06$ mol dm^{-3}
pH = −log 0.06 = 1.2

4. C
$[H^+(aq)]$ in the solution of acid *J* $= 10^{-1}$ M = 0.1 M
$[H^+(aq)]$ in the solution of acid *K* $= 10^{-2}$ M = 0.01 M

5. D
For the 0.064 M H*X*(aq), pH = −log 0.064 = 1.2
For the 0.0032 M H*X*(aq), pH = −log 0.0032 = 2.5

6. B
$HNO_3(aq) \longrightarrow H^+(aq) + NO_3^-(aq)$
pH of the sample of $HNO_3(aq) = 1.8 = -\log[H^+(aq)]$
$[H^+(aq)]$ in the sample of $HNO_3(aq) = 10^{-1.8}$ M
When 200 cm^3 of water is mixed with 100 cm^3 of the sample, the volume of the resultant mixture becomes 300 cm^3.
Let the $[H^+(aq)]$ in the resultant mixture be M_2.
$10^{-1.8} \times \frac{100}{1000} = M_2 \times \frac{300}{1000}$
$M_2 = 5.3 \times 10^{-3}$
∴ pH of the resultant mixture $= -\log 5.3 \times 10^{-3}$
$= 2.3$

7. C
Volume of the resultant solution $= (10 + 990) \text{ cm}^3$
$= 1000 \text{ cm}^3$
As the concentration of $H^+(aq)$ in the sample of HCl(aq) decreases by a factor of 100 after dilution, the pH of resultant solution is 2 + 2 = 4.

8. C
Number of moles of $H^+(aq)$ from H*A*
$= 6.0 \times \frac{25.0}{1000}$ mol = 0.15 mol
Number of moles of $H^+(aq)$ from H*B*
$= 3.0 \times \frac{45.0}{1000}$ mol = 0.14 mol
Concentration of $H^+(aq)$ in the resulting solution
$= \frac{(0.15 + 0.14) \text{ mol}}{(\frac{25.0 + 45.0}{1000}) \text{ dm}^3} = 4.1 \text{ mol dm}^{-3}$

9. B
pH of 0.1 M $HNO_3(aq) = -\log 0.1 = 1$

10. B

B. Structured questions (p.90)

11. (a) pH of 0.0005 M HCl(aq)
$= -\log 0.0005 = 3.3$ 1
Hence, the colour of methyl red is red. 1
(b) pH of the solution $= -\log 3.2 \times 10^{-13} = 12.5$ 1
Hence, the colour of phenolphthalein is pink. 1
(c) Around 4.4 to 6.0 1

12. (a) pH meter 1

(b) The solution is acidic. 1

(c) $2.87 = -\log[H^+(aq)]$

$[H^+(aq)] = 10^{-2.87}$ mol dm^{-3}

$= 1.35 \times 10^{-3}$ mol dm^{-3} 1

(d) $pH = -\log(\frac{1.35 \times 10^{-3}}{2}) = 3.17$ 1

(e) Easy to use 1

Quick to obtain accurate readings 1

Chapter 17

A. Multiple-choice questions (p.106)

1. A

Sulphurous acid is a weak and dibasic acid. It ionizes slightly in water according to the following equation:

$H_2SO_3(l) \overset{water}{\rightleftharpoons} H^+(aq) + HSO_3^-(aq)$

$HSO_3^-(aq) \rightleftharpoons H^+(aq) + SO_3^{2-}(aq)$

As the ionization does not occur completely, the concentration of $H^+(aq)$ in 1.0 M $H_2SO_3(aq)$ is smaller than 2.0 M.

2. D

Dilute nitric acid reacts with sodium carbonate according to the following equation:

$2HNO_3(aq) + Na_2CO_3(aq)$

$\longrightarrow 2NaNO_3(aq) + CO_2(g) + H_2O(l)$

3. C

Both NaOH(aq) and $NH_3(aq)$ form a white precipitate of $Mg(OH)_2$ with $Mg(NO_3)_2(aq)$.

4. B

Carbonic acid is a weak acid while sulphuric acid is a strong acid. Sulphuric acid has a higher concentration of $H^+(aq)$ ions than carbonic acid for the same molar concentration.

5. C

6. A

Aqueous ammonia ionizes slightly in water while hydrochloric acid, sodium chloride and sodium hydroxide ionize or dissociate completely in water.

7. A

8. D

9. C

A weak acid ionizes only slightly in water to give $H^+(aq)$ and its corresponding anion. Hence, the pH of this solution is greater than 2.0. In addition, this solution has a rather low concentration of mobile ions. Hence, it has poor electrical conductivity.

10. C (77%)

Option (A): HCl(aq) is a stronger acid than $CH_3COOH(aq)$, irrespective of concentration. *X* has a higher pH than *Z* because the $[H^+(aq)]$ in *X* is lower than that in *Z*.

Option (B): the pH of a solution is independent of the volume of the solution. *Y* has a lower pH than *X* because *Y* has a higher $[H^+(aq)]$ than *X*.

Option (D): *Y* has a lower pH than *Z* because the $[H^+(aq)]$ in *Y* is higher than that in *Z*.

B. Structured questions (p.107)

11. (a) $[H^+(aq)] = 10^{-2.0}$ M = 0.01 M

As *Y* is a strong and monobasic acid, 1 mol of *Y* ionizes completely or almost completely in water to give 1 mol of $H^+(aq)$ ions.

∴ the molarity of *Y* in the solution is 0.01 M. 1

(b) *X* and *Z* have the same molarity and basicity but *X* has a lower pH than *Z*. This suggests that *X* ionizes more completely in water than *Z*. 1

Hence, *X* is a stronger acid than *Z*. 1

12. (a) $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ 1

(b) Measure the pH/electrical conductivity of the two aqueous solutions of alkalis (i.e. aqueous ammonia and sodium hydroxide solution) of the same molar concentration. 1

The pH/electrical conductivity of aqueous ammonia is lower than that of sodium hydroxide solution. 1

This is because aqueous ammonia is a weak alkali while sodium hydroxide solution is a strong alkali. For the same molar concentration, aqueous ammonia has a lower concentration of $OH^-(aq)$ ions/ mobile ions compared with sodium hydroxide solution. 1

13. (a) Prepare two lumps of magnesium carbonate with very similar sizes and masses. Add one to each of the acids (i.e. sulphuric acid and sulphurous acid) with the same molar concentration. 1

Compare the rate of evolution of the colourless gas bubbles/vigour of the two reactions. 1

(b) Sulphuric acid shows a higher rate of evolution of colourless gas bubbles/ reacts more vigorously with magnesium carbonate compared with sulphurous acid. 1

This is because sulphuric acid is a strong acid while sulphurous acid is a weak acid. For the same molar concentration, sulphuric acid has a higher concentration of $H^+(aq)$ ions for the reaction with magnesium carbonate compared with sulphurous acid. 1

Chapter 18

A. Multiple-choice questions (p.146)

1. A

Equation of the reaction involved: $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

	Number of moles of $H^+(aq)$ reacted / mol	Number of moles of $OH^-(aq)$ reacted / mol	Volume of the reaction mixture / cm^3
(1)	$1.0 \times \frac{25.0}{1000} = 0.025$		25.0 + 25.0 = 50.0
(2)	$1.0 \times \frac{100.0}{1000} = 0.1$		100.0 + 100.0 = 200.0
(3)	$2.0 \times \frac{50.0}{1000} = 0.1$		50.0 + 50.0 = 100.0

In (2), although the volumes of the solutions used are four times that used in (1), the number of moles of $H^+(aq)$ ions and $OH^-(aq)$ ions reacting is also four times that in (1). Four times as much heat is given out, but this is used to heat up four times the volume of the reaction mixture. Hence, the maximum temperature rise in (1) and (2) are about the same.

In (3), the number of moles of $H^+(aq)$ ions and $OH^-(aq)$ ions reacting is also four times that in (1), but the heat produced is used to heat up only twice the volume of the reaction mixture. Hence, the maximum temperature rise in (3) is higher than that in (1).

2. A (66%)

W, *X*, *Y* and *Z* represent $MgCl_2(aq)$, $NaOH(aq)$, $Na_2CO_3(aq)$ and $HCl(aq)$ respectively.

Mixing *W* and *X*:

$MgCl_2(aq) + 2NaOH(aq) \longrightarrow 2NaCl(aq) + Mg(OH)_2(s)$ white precipitate

Mixing *W* and *Y*:

$MgCl_2(aq) + Na_2CO_3(aq) \longrightarrow 2NaCl(aq) + MgCO_3(s)$ white precipitate

Mixing *W* and *Z*:

$MgCl_2(aq) + HCl(aq) \longrightarrow$ no reaction

3. B

Option (A): no displacement reaction occurs as zinc is less reactive than magnesium.

Option (B): neutralization reaction occurs and it can be represented by the equation: $Zn(OH)_2(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + 2H_2O(l)$

Option (C): magnesium chloride is a salt. It has no reaction with a metal oxide.

Option (D): zinc chloride is a water soluble salt. It cannot be prepared by precipitation.

4. C

There is no reaction between zinc nitrate and dilute hydrochloric acid.

5. B

The equation for the reaction involved in (2) is $Na_2SO_4(aq) + CaCl_2(aq) \longrightarrow CaSO_4(s) + 2NaCl(aq)$.

6. D

Upon mixing $Ca(OH)_2(aq)$ and $H_2SO_4(aq)$, the following precipitation reaction occurs: $Ca(OH)_2(aq) + H_2SO_4(aq) \longrightarrow CaSO_4(s) + 2H_2O(l)$. $CaSO_4$ is insoluble in water. The concentration of mobile ions is the lowest in the resultant mixture in option (D).

7. B

When aqueous ammonia is added to an aqueous solution of calcium ions, no precipitate would form. This is because the calcium hydroxide formed is soluble in aqueous ammonia. Besides, neither calcium nitrate solution nor magnesium nitrate solution reacts with dilute hydrochloric acid.

8. D

9. C

The (dirty) green precipitate and white precipitate produced in these reactions are $Fe(OH)_2(s)$ and $BaSO_4(s)$ respectively.

10. B

The reactions involved can be represented by the following equations:

$PbCO_3(s) + 2HNO_3(aq) \longrightarrow Pb(NO_3)_2(aq) + CO_2(g) + H_2O(l)$

$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow PbSO_4(s) + 2NaNO_3(aq)$

B. Structured questions (p.147)

11. (a) Zinc nitrate 1

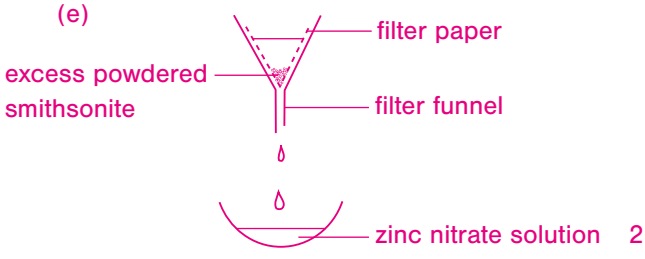
(b) $ZnCO_3(s) + 2HNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + CO_2(g) + H_2O(l)$ 1

(c) Effervescence stops/colourless gas bubbles stop evolving. 1

Some powdered smithsonite remains undissolved in the acid. 1

(d) This is to ensure that all the dilute nitric acid has reacted and the product is not contaminated by the acid. 1

(e)



2

(1 mark for correct drawing; 1 mark for correct labelling)

(f) Leave the hot saturated solution to cool slowly to room temperature. Zinc nitrate crystals will form after some time. 1

Filter the residual solution to obtain the crystals. Wash the crystals with a little cold distilled water. Finally, dry the crystals using filter paper. 1

12. (a) (i) *x*: NaOH(aq) 1

y: $Zn(OH)_2(s)$ 1

(ii) By filtration 1

(b) Dilute HCl(aq) 1

Chapter 19

A. Multiple-choice questions (p.195)

1. D (53%)
2. C

$H_2SO_4(aq) + 2KOH(aq) \longrightarrow K_2SO_4(aq) + 2H_2O(l)$

Number of moles of H_2SO_4

$= 0.5 \times \frac{20.0}{1000}$ mol = 0.01 mol

To give a resultant solution of pH 7, the sulphuric acid should be completely neutralized by potassium hydroxide solution. From the equation, mole ratio of H_2SO_4 to KOH = 1 : 2. Hence, the number of moles of KOH required = 0.01 mol × 2 = 0.02 mol

Option (A): number of moles of KOH

$= 0.5 \times \frac{20.0}{1000}$ mol = 0.01 mol

Option (B): number of moles of KOH

$= 0.5 \times \frac{10.0}{1000}$ mol = 0.005 mol

Option (C): number of moles of KOH

$= 1.0 \times \frac{20.0}{1000}$ mol = 0.02 mol

Option (D): number of moles of KOH

$= 1.0 \times \frac{40.0}{1000}$ mol = 0.04 mol

3. D

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

Number of moles of HCl

$= 0.10 \times \frac{20}{1000} = 0.002$ mol

Number of moles of NaOH

$= 0.10 \times \frac{10}{1000} = 0.001$ mol

From the equation, 0.001 mol of NaOH requires only 0.001 mol of HCl for complete neutralization. Hence, HCl is in excess.

Concentration of $H^+(aq)$ ions in the resulting mixture

$= \frac{0.002 - 0.001}{\frac{10 + 20}{1000}}$ mol dm^{-3}

pH of the resulting mixture

$= -\log \left(\frac{0.002 - 0.001}{\frac{10 + 20}{1000}}\right) = 1.48$

4. D

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

From the equation, mole ratio of NaOH to HCl = 1 : 1.

∴ concentration of HCl(aq)

$= \frac{0.245 \times \frac{25.0}{1000}}{\frac{13.49}{1000}}$ mol dm^{-3} = 0.454 mol dm^{-3}

5. C

Option (A): if the pipette is rinsed with distilled water only, the concentration of sodium hydroxide solution in it may be smaller than 0.245 M. As a result, a smaller volume of hydrochloric acid may be needed for neutralizing all the sodium hydroxide solution.

Option (C): if the conical flask is rinsed with sodium hydroxide solution, the volume of 0.245 M sodium hydroxide solution in it may be greater than 25.0 cm^3. As a result, a greater volume of hydrochloric acid may be needed for neutralizing all the sodium hydroxide solution.

Option (D): if the filter funnel is left on the top of the burette, hydrochloric acid clinging onto the stem of the funnel may fall into the burette. As a result, a smaller volume of hydrochloric acid would be needed for neutralizing all the sodium hydroxide solution.

6. C
7. B

$H_2SO_4(aq) + X_2CO_3(aq) \longrightarrow X_2SO_4(aq) + CO_2(g) + H_2O(l)$

From the equation, mole ratio of H_2SO_4 to X_2CO_3 = 1 : 1.

Let the relative atomic mass of X be a.

Number of moles of X_2CO_3

$$= \frac{0.53}{2a + 12.0 + 16.0 \times 3} = 0.2 \times \frac{25.00}{1000}$$

$$= \frac{0.53}{2a + 60.0} = 5 \times 10^{-3}$$

$a = 23.0$

$\therefore$ the relative atomic mass of X is 23.0.

8. A

$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$

Number of moles of HCl used

$$= 0.10 \times \frac{23.50}{1000} \text{ mol} = 2.35 \times 10^{-3} \text{ mol}$$

From the equation, mole ratio of HCl to Na_2CO_3 = 2 : 1.

$\therefore$ number of moles of Na_2CO_3 in the 250.0 cm^3 solution

$$= 2.35 \times 10^{-3} \times \frac{1}{2} \times \frac{250.0}{25.0} \text{ mol}$$

$$= 0.01175 \text{ mol}$$

Number of moles of $Na_2CO_3 \cdot nH_2O$

$$= 0.01175 = \frac{1.46}{23.0 \times 2 + 12.0 + 16.0 \times 3 + (1.0 \times 2 + 16.0) \times n}$$

$$= 0.01175 = \frac{1.46}{106.0 + 18n}$$

$n = 1.01$

9. C

10. C

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

From the equation, mole ratio of NaOH to HCl = 1 : 1.

$\therefore$ number of moles of HCl reacted with $CaCO_3$ in the egg shell

$$= (1.00 \times \frac{50.0}{1000} - 0.500 \times \frac{22.65}{1000}) \text{ mol}$$

$$= 0.0387 \text{ mol}$$

$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

From the equation, mole ratio of HCl to $CaCO_3$ = 2 : 1.

$\therefore$ percentage by mass of $CaCO_3$ in the egg shell

$$= \frac{0.0387 \times \frac{1}{2} \times (40.1 + 12.0 + 16.0 \times 3) \text{ g}}{3.85 \text{ g}} \times 100\%$$

$$= 50.3\%$$

B. Structured questions (p.196)

11. (a) Using a clean 25.0 cm^3 pipette, transfer 25.0 cm^3 of 0.25 M solution of acid X to a clean 250.0 cm^3 volumetric flask. 1

Add distilled water up to the graduation mark of the flask. Finally, stopper the flask. Invert it several times to mix the contents well. 1

(b) From yellow to orange 1

(c) Number of moles of NaOH

$$= 0.05 \times \frac{25.0}{1000} \text{ mol} = 1.25 \times 10^{-3} \text{ mol}$$

Number of moles of acid X 1

$$= \frac{0.25}{10} \times \frac{24.70}{1000} \text{ mol}$$

$$= 6.175 \times 10^{-4} \text{ mol}$$

$$\frac{\text{Number of moles of NaOH}}{\text{Number of moles of acid } X} = \frac{1.25 \times 10^{-3}}{6.175 \times 10^{-4}}$$

$$= 2.02 \quad 1$$

$\therefore$ the basicity of acid X is 2.

(d) Measure the pH of the reaction mixture after each addition of a small but fixed volume of diluted solution of acid X to the standard sodium hydroxide solution by using a pH meter/data-logger with a pH sensor. 1

Using the pH readings obtained, plot a titration curve of this titration. The point where there is a sharp change in pH is the equivalence point of this titration. 1

OR

Measure the temperature of the reaction mixture after each addition of a small but fixed volume of diluted solution of acid X to the standard sodium hydroxide solution by using a thermometer. (1)

Using the temperature readings obtained, plot a graph of the temperature of the reaction mixture against the volume of the diluted solution of acid X added. The point where the temperature of the reaction mixture reaches the maximum (determined by extrapolation) is the equivalence point of this titration. (1)

12. (a) Aqueous ammonia 1

$NH_3(aq) + HClO_4(aq) \longrightarrow NH_4ClO_4(aq)$ 1

(b) (i) Methyl orange 1

(ii) From the equation in (a), mole ratio of $HClO_4$ to NH_3 = 1 : 1.

$\therefore$ molarity of the $NH_3(aq)$

$$= \frac{0.65 \times \frac{26.90}{1000}}{\frac{25.0}{1000}} \text{ mol dm}^{-3} \quad 1$$

$$= 0.70 \text{ mol dm}^{-3} \quad 1$$

(c) (i) From the equation in (a), mole ratio of $HClO_4$ to NH_4ClO_4 = 1 : 1.

$\therefore$ theoretical yield of NH_4ClO_4

$$= 0.65 \times \frac{26.90}{1000} \times 117.5 \text{ g} \quad 1$$

$$= 2.05 \text{ g} \quad 1$$

(Note: the formula mass of NH_4ClO_4 is 117.5 g mol^{-1})

(ii) The actual yield is much lower than the theoretical yield probably because:

- not all NH_4ClO_4 was crystallized out from the saturated NH_4ClO_4 solution 1
- some NH_4ClO_4 crystals were lost during filtering and drying (1)

13. HKDSE 2018 Paper 1B Q7

14. (a) (i) $SO_2(g) + H_2O_2(aq) \longrightarrow H_2SO_4(aq)$ 1

(ii) $H_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$ 1

(b) Reasonable average volume of NaOH(aq) used

$= \frac{6.20 + 6.25 + 6.30}{3}\ cm^3 = 6.25\ cm^3$

From the equations in (a), mole ratio of NaOH to H_2SO_4 to SO_2 = 2 : 1 : 1.

∴ number of moles of SO_2 in the dried fruit sample

$= 0.010 \times \frac{6.25}{1000} \times \frac{1}{2}$ mol 1

$= 3.13 \times 10^{-5}$ mol 1

(c) Mass of SO_2 in 25.0 g of the dried fruit sample

$= 3.13 \times 10^{-5}$ mol $\times$ (32.1 + 16.0 $\times$ 2) g $mol^{-1} \times \frac{1000\ mg}{1\ g}$

= 2.00 mg 1

Mass of SO_2 per kg of the dried fruit sample

$= \frac{2.00\ mg}{\frac{25.0}{1000}\ kg} = 80.0\ mg\ kg^{-1}$ 1

As the calculated value is smaller than 100 mg kg^{-1}, the dried fruit sample has not exceeded the limit.

15. (a) $CaCO_3(s) + 2HNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + H_2O(l) + CO_2(g)$ 1

(b) From colourless to pink 1

(c) (i) $HNO_3(aq) + NaOH(aq) \longrightarrow NaNO_3(aq) + H_2O(l)$

From the equation, mole ratio of NaOH to HNO_3 = 1 : 1.

∴ number of moles of unreacted HNO_3

$= 0.40 \times \frac{20.00}{1000}$ mol $= 8.0 \times 10^{-3}$ mol 1

(ii) From the equation in (a), mole ratio of HNO_3 to $CaCO_3$ = 2 : 1.

∴ number of moles of $CaCO_3$ in the limestone sample

$= (0.50 \times \frac{50.0}{1000} - 8.0 \times 10^{-3}) \times \frac{1}{2}$ mol

$= 8.5 \times 10^{-3}$ mol 1

(iii) Percentage by mass of $CaCO_3$ in the limestone sample

$= \frac{8.5 \times 10^{-3} \times (40.1 + 12.0 + 16.0 \times 3)\ g}{1.28\ g} \times 100\%$

= 66.5% 1

16. (a) $Fe_2O_3 \cdot nH_2O(s) + 6HCl(aq) \longrightarrow 2FeCl_3(aq) + (3 + n)H_2O(l)$ 1

(b) $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

From the equation, mole ratio of NaOH to HCl = 1 : 1.

∴ number of moles of HCl reacted with the $Fe_2O_3 \cdot nH_2O$ in the rust sample

$= (1.50 \times \frac{40.0}{1000} - 0.50 \times \frac{35.16}{1000})$ mol 1

= 0.0424 mol 1

(c) From the equation in (a), mole ratio of HCl to $Fe_2O_3 \cdot nH_2O$ = 6 : 1.

∴ number of moles of $Fe_2O_3 \cdot nH_2O$ in the rust sample

$= \frac{0.0424}{6} = \frac{1.89}{(55.8 \times 2 + 16.0 \times 3) + (1.0 \times 2 + 16.0) \times n}$ 1

$n = 5.99 \approx 6$ 1

(d) The number of water of crystallization per formula unit of hydrated iron(III) oxide depends on the amount of water reacted with an iron sample in the formation of rust in each case. 1

17. (a) In step (2), the flask should not be fully filled with distilled water. The correct procedure is:

Fill the flask with more distilled water until the bottom of the meniscus reaches the graduation mark. Stopper and invert the flask several times. 1

In step (3), the burette should also be rinsed with the given standard hydrochloric acid. The correct procedure is:

Fill a burette, which has been well rinsed with distilled water followed by the given standard hydrochloric acid, with the standard hydrochloric acid and make sure that the jet is filled completely. 1

In step (4), the 25.0 cm^3 diluted sample should not be transferred to a 250 cm^3 beaker. The correct procedure is:

Using a cleaned pipette, transfer 25.0 cm^3 of the diluted sample to a cleaned conical flask and add a few drops of methyl orange indicator. 1

(b) Step (1): 25.0 cm^3 of a drain cleaner was transferred to a 250.0 cm^3 volumetric flask using a cleaned pipette. 1

Step (2): The flask was filled with distilled water until the bottom of the meniscus reached the graduation mark. The flask was stoppered and inverted several times. 1

Answers to Part exercise

Chapter 19

A. Multiple-choice questions (p.200)

1. D

The equation of the reaction is:

$CuCO_3(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + CO_2(g) + H_2O(l)$

2. D

For option (D), the equation of the reaction is:

$Ca(OH)_2(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + 2H_2O(l)$

3. D

Aqueous ammonia is an alkali, but it does not contain metal ions.

4. D

Statement (1): when the concentration of $OH^-(aq)$ of a solution decreases, the pH of the solution also decreases.

Statement (3): sodium hydroxide solution is a strong alkali. It dissociates completely in water regardless of concentration.

5. C

(1): adding sodium chloride solution to 25.0 cm^3 of 0.5 M hydrochloric acid will dilute the acid. This lowers the concentration of $H^+(aq)$ ions in the acid. As a result, the pH of the acid will increase.

(2): the resultant solution will be 50.0 cm^3 of 0.5 M hydrochloric acid.

(3): ethanoic acid is a weak acid, which ionizes only slightly in water to give $H^+(aq)$ ions. Hence, adding the ethanoic acid to 25.0 cm^3 of 0.5 M hydrochloric acid will lower the concentration of $H^+(aq)$ ions in the acid. As a result, the pH of the acid will increase.

6. A (85%)

The pH of vinegar is around 3.

7. C

The volume of the resultant mixture is the same in all the four combinations. But the number of moles of $H^+(aq)$ ions and $OH^-(aq)$ ions reacting is the largest in the combination for option (C). This combination would give out the most heat, and hence would produce the greatest temperature rise.

8. C (68%)

(1): neither $AgNO_3(aq)$ nor $NaNO_3(aq)$ reacts with $Cu(NO_3)_2(aq)$.

(2): only $AgNO_3(aq)$ reacts with HCl(aq) to give a white precipitate (AgCl(s)).

(3): only $AgNO_3(aq)$ reacts with KOH(aq) to give a dark brown precipitate ($Ag_2O(s)$).

9. A

$Pb(NO_3)_2(aq) + Na_2SO_4(aq) \longrightarrow PbSO_4(s) + 2NaNO_3(aq)$

Number of moles of Na_2SO_4

$= 0.1 \times \frac{20.0}{1000}$ mol = 0.002 mol

From the equation, mole ratio of Na_2SO_4 to $PbSO_4$ = 1 : 1.

$\therefore$ mass of $PbSO_4(s)$ formed

$= 0.002 \times (207.2 + 32.1 + 16.0 \times 4)$ g = 0.607 g

10. A (59%)

$Pb(NO_3)_2(aq) + 2NaCl(aq) \longrightarrow PbCl_2(s) + 2NaNO_3(aq)$

Number of moles of $Pb(NO_3)_2$

$= 0.50 \times \frac{25.00}{1000}$ mol = 0.0125 mol

Number of moles of NaCl

$= 1.00 \times \frac{50.00}{1000}$ mol = 0.0500 mol

0.0125 mol of $Pb(NO_3)_2$ requires only 0.0125 mol $\times$ 2 = 0.0250 mol of NaCl for complete reaction. Hence, $Pb(NO_3)_2$ is the limiting reactant.

Concentration of $Cl^-(aq)$ in the mixture

$= \frac{0.0500 - 0.0125 \times 2}{\frac{25.00 + 50.00}{1000}}$ M = 0.33 M

11. A (57%)

$3CaCl_2(aq) + 2Na_3PO_4(aq) \longrightarrow Ca_3(PO_4)_2(s) + 6NaCl(aq)$

Number of moles of $CaCl_2$

$= 0.30 \times \frac{100.0}{1000}$ mol = 0.030 mol

Number of moles of Na_3PO_4

$= 0.10 \times \frac{300.0}{1000}$ mol = 0.030 mol

0.030 mol of $CaCl_2$ requires only 0.030 mol $\times \frac{2}{3}$ = 0.020 mol of Na_3PO_4 for complete reaction. Hence, $CaCl_2$ is the limiting reactant.

From the equation, mole ratio of $CaCl_2$ to $Ca_3(PO_4)_2$ = 3 : 1.

∴ number of moles of $Ca_3(PO_4)_2$ formed

$= \frac{0.030}{3}$ mol = 0.010 mol

12. D

$Pb(OH)_2(s) \xrightarrow{\text{dilute } HNO_3(aq)} Pb(NO_3)_2(aq)$

$\xrightarrow{NaCl(aq)} PbCl_2(s)$

13. A

14. A (35%)

$3KOH(aq) + H_3A(aq) \longrightarrow K_3A(aq) + 3H_2O(l)$

3 mol 1 mol

The acid is a tribasic acid.

15. D (50%)

The mixture of *W* and *X* is a mixture of two strong monobasic acids of 0.20 M. Hence, the concentration of $H^+(aq)$ ions in the mixture is 0.20 M.

The mixture of *X* and *Y* is a mixture of a strong monobasic acid of 0.20 M and a weak monobasic acid of 0.20 M. Hence, the concentration of $H^+(aq)$ ions in the mixture is smaller than 0.20 M.

Therefore, the pH of the mixture of *W* and *X* is lower than that of the mixture of *X* and *Y*.

16. D (51%)

In the absence of water, there is no reaction between acidic gases and CaO(s).

17. C

18. C (48%)

$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

As the $OH^-(aq)$ ions have reacted, more NH_3 molecules will ionize in water to give $OH^-(aq)$ ions for the reaction with HCl(aq). Hence, 1 mol of $NH_3(aq)$ can completely neutralize 1 mol of HCl(aq).

B. Structured questions (p.202)

19. (a) 2

	H	N	S	O
Mass / g	3.1	14.4	33.1	49.4
Number of moles of atoms / mol	$\frac{3.1}{1.0} = 3.10$	$\frac{14.4}{14.0} = 1.03$	$\frac{33.1}{32.1} = 1.03$	$\frac{49.4}{16.0} = 3.09$
Simplest whole number mole ratio of atoms	$\frac{3.10}{1.03} = 3.01 \approx 3$	$\frac{1.03}{1.03} = 1$	$\frac{1.03}{1.03} = 1$	$\frac{3.09}{1.03} = 3$

∴ the empirical formula of sulphamic acid is H_3NSO_3.

(1 mark for correct calculations of the number of moles of atoms; 1 mark for the correct calculations of the simplest whole number mole ratio of atoms)

(b) Let the molecular formula of sulphamic acid be $(H_3NSO_3)_n$, where *n* is an integer.

$n \times (1.0 \times 3 + 14.0 + 32.1 + 16.0 \times 3) = 97.1$ 1

$n = 1$

∴ the molecular formula of sulphamic acid is H_3NSO_3.

(c) Mass of sulphamic acid in the cleaning powder = 5.00 g × 14.5% = 0.725 g

Molarity of the solution prepared

$= \frac{\frac{0.725}{97.1}}{\frac{200}{1000}}$ mol dm^{-3} 1

= 0.0373 mol dm^{-3} 1

20. HKDSE 2019 Paper 1B Q4

21. (a) When H*X* is diluted 10 times, the pH of its aqueous solution increases by one unit. 1

This suggests that H*X* ionizes completely in water and so the concentration of $H^+(aq)$ ions decreases by 10 times. 1

Hence, H*X* is a strong acid.

(b) $x = 10^{-0.92}$ M = 0.12 M 1

(c) Using a clean pipette, transfer 25.0 cm^3 of H*X* to a clean 250.0 cm^3 volumetric flask. 1

Add distilled water up to the graduation mark of the flask. Finally stopper the flask, invert it several times to mix the contents well. 1

22. (a) Monobasic acid is an acid which produces one hydrogen ion when one molecule of the acid ionizes completely in water. 1

(b) (i) Baking powder is alkaline. It can neutralize the methanoic acid injected into the skin. 1

$H^+(aq) + HCO_3^-(aq) \longrightarrow H_2O(l) + CO_2(g)$ 1

(ii) From the equation in (i), mole ratio of HCOOH to $NaHCO_3$ = 1 : 1.

∴ mass of $NaHCO_3$ required

$= 1.50 \times 10^{-4} \times (23.0 + 1.0 + 12.0 + 16.0 \times 3)$ g 1

= 0.0126 g 1

(iii) Sodium hydroxide solution is corrosive. It can attack the skin. 1

23. (a) A weak acid is an acid which ionizes slightly in water. 1

(b) $H_3A(aq) + 3NaOH(aq) \longrightarrow Na_3A(aq) + 3H_2O(l)$ 1

(c) Number of moles of NaOH used

$= 0.2 \times \frac{1.30}{1000}$ mol $= 2.6 \times 10^{-4}$ mol 1

From the equation in (b), mole ratio of NaOH to H_3A = 3 : 1.

∴ number of moles of $H_3A = \frac{2.6 \times 10^{-4}}{3}$ mol

Concentration of citric acid in the lemon juice

$= \frac{\frac{2.6 \times 10^{-4}}{3} \text{ mol}}{\frac{5.0}{1000} \text{ dm}^3} \times 192.0 \text{ g mol}^{-1}$

$= 3.3 \text{ g dm}^{-3}$ 1

(d) The lemon juice does not contain any substance that reacts with sodium hydroxide solution. 1

(e) Any one of the following:

This saves chemicals. 1

This saves time on carrying out the experiment. (1)

This minimizes the potential hazards of the experiment. (1)

This reduces chemical wastes produced. (1)

24. (a) This is to save sodium hydroxide solution. 1

(b) Phenolphthalein 1

From colourless to pink 1

(c) Reasonable average volume of NaOH(aq) used

$= \frac{22.50 + 22.40 + 22.60}{3} \text{ cm}^3 = 22.50 \text{ cm}^3$

$CH_3COOH(aq) + NaOH(aq) \longrightarrow CH_3COONa(aq) + H_2O(l)$

Number of moles of NaOH used

$= 0.050 \times \frac{22.50}{1000}$ mol $= 1.1 \times 10^{-3}$ mol 1

From the equation, mole ratio of NaOH to CH_3COOH = 1 : 1.

∴ number of moles of CH_3COOH in 25.0 cm^3 of the diluted vinegar solution = 1.1×10^{-3} mol

Number of moles of CH_3COOH in the 10.0 cm^3 vinegar

$= 1.1 \times 10^{-3} \text{ mol} \times \frac{250.0}{25.0} = 1.1 \times 10^{-2}$ mol 1

Concentration of CH_3COOH in this brand of vinegar

$= \frac{1.1 \times 10^{-2}}{\frac{10.0}{1000}} \text{ mol dm}^{-3} = 1.1 \text{ mol dm}^{-3}$ 1

(d) (i) The sodium hydroxide solution clinging onto the stem of the funnel may fall into the burette. As a result, a smaller volume of sodium hydroxide solution would be needed for neutralizing all the diluted vinegar. 1

(ii) After washing, some diluted vinegar may remain in the flask. As a result, a larger volume of sodium hydroxide solution may be needed for neutralizing all the diluted vinegar. 1

25. (a) A white precipitate forms. 1

(b) $Ba^{2+}(aq) + SO_4^{2-}(aq) \longrightarrow BaSO_4(s)$ 1

(c) (i) The electrical conductivity of the reaction mixture decreases. This is because $BaSO_4(s)$ and $H_2O(l)$ form when dilute $H_2SO_4(aq)$ is added to $Ba(OH)_2(aq)$. This lowers the concentration of mobile ions in the reaction mixture. 1

(ii) The electrical conductivity of the reaction mixture increases. This is because the concentration of mobile ions in the mixture increases when dilute $H_2SO_4(aq)$ is added in excess. 1

(d) 33.0 cm^3 1

(e) Number of moles of $Ba(OH)_2$

$= 0.10 \times \frac{25.0}{1000}$ mol $= 2.5 \times 10^{-3}$ mol 1

From the equation in (b), mole ratio of Ba^{2+} to SO_4^{2-} = 1 : 1.

∴ number of moles of $H_2SO_4 = 2.5 \times 10^{-3}$ mol

Molarity of the $H_2SO_4(aq)$

$= \frac{2.5 \times 10^{-3}}{\frac{33.0}{1000}} \text{ mol dm}^{-3} = 0.076 \text{ mol dm}^{-3}$ 1

26. HKDSE 2020 Paper 1B Q4

27. HKDSE 2020 Paper 1B Q2

28. HKDSE 2021 Paper 1B Q7

Answers to Revision test

Chapter 19

A. Multiple-choice questions (p.208)

1. B

2. D

X has a lower pH than *Y*. As pH = $-\log[H^+(aq)]$, the concentration of $H^+(aq)$ in *X*(aq) is higher than that in *Y*(aq).

3. A

In 100.0 cm^3 aqueous solution, 0.1 mol of H_2SO_4 ionizes almost completely to form more than 2 mol of mobile ions, while 0.1 mol of KOH dissociates completely to form 2 mol of mobile ions. $NH_3(aq)$ and $CH_3COOH(aq)$ are weak alkali and weak acid respectively. Only a small proportion of their own molecules ionize in water to give mobile ions.

4. B

Reaction between Solution *P* and NaOH(aq):

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

0.005 mol ?

Reaction between Solution *Q* and NaOH(aq):

$CH_3COOH(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

0.005 mol ?

According to the equations, the HCl(aq) and the $CH_3COOH(aq)$ require the same number of moles of NaOH(aq) for complete neutralization.

5. C

$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$

Experiment	Number of moles of HCl(aq) / mol	Number of moles of NaOH(aq) / mol	Number of moles of $H_2O(l)$ formed / mol
I	0.025	0.025	0.025
II	0.05 (in excess)	0.025	0.025

Since the numbers of moles of HCl(aq) and NaOH(aq) reacting are the same in both experiments I and II, the amount of heat produced in both experiments is equal.

The volume of the mixture in experiment I is smaller than that in experiment II. As a smaller volume of mixture is heated by equal amount of heat, the maximum temperature rise of the mixture in experiment I is higher than that in experiment II.

6. C

(1): $Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$

(3): $Ca(NO_3)_2(aq) + 2NaOH(aq) \longrightarrow Ca(OH)_2(s) + 2NaNO_3(aq)$

7. C

Lead(II) chloride is an insoluble salt. It could be prepared by precipitation.

$Pb(NO_3)_2(aq) + 2NaCl(aq) \longrightarrow PbCl_2(s) + 2NaNO_3(aq)$

8. D

Mass of $Ni(NO_3)_2 \cdot 6H_2O$ required

$= 0.1 \times \frac{250.0}{1000} \times [58.7 + (14.0 + 16.0 \times 3) \times 2 + (1.0 \times 2 + 16.0) \times 6]$ g

$= 7.27$ g

9. C

$Fe_2O_3(s) + 3H_2SO_4(aq) \longrightarrow Fe_2(SO_4)_3(aq) + 3H_2O(l)$

From the equation, mole ratio of Fe_2O_3 to H_2SO_4 = 1 : 3.

$\therefore$ volume of $H_2SO_4(aq)$ required

$= \frac{\frac{2.71}{55.8 \times 2 + 16.0 \times 3} \times 3}{0.20}$ dm^3

$= 0.255$ dm^3 = 255 cm^3

10. B

$2MHCO_3(aq) + H_2SO_4(aq) \longrightarrow M_2SO_4(aq) + 2CO_2(g) + 2H_2O(l)$

Number of moles of H_2SO_4 used

$= 0.5 \times \frac{20.0}{1000}$ mol = 0.01 mol

From the equation, mole ratio of H_2SO_4 to $MHCO_3$ = 1 : 2.

$\therefore$ number of moles of $MHCO_3$ reacted

= 0.01 mol × 2 = 0.02 mol

Let the relative atomic mass of *M* be *a*.

$0.02 = \frac{2.00}{a + 1.0 + 12.0 + 16.0 \times 3}$

$a = 39.0$

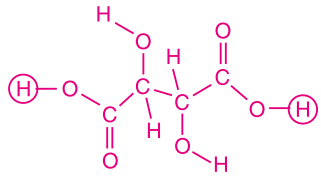
$\therefore$ the relative atomic mass of *M* is 39.0.

11. D

Concentrated ethanoic acid is a weak acid. Concentrated hydrochloric acid does not attack copper.

B. Structured questions (p.209)

12. (a)



1

(b) 2 1

(c) (i) Effervescence occurs/colourless gas bubbles evolve. 1

(ii) When dissolved in water, tartaric acid ionizes to give $H^+(aq)$ which react with $HCO_3^-(aq)$ to give $CO_2(g)$. 1

$H^+(aq) + HCO_3^-(aq) \longrightarrow CO_2(g) + H_2O(l)$ 1

13. (a) Add a few drops of the milk sample and lemon juice sample to a piece of pH paper respectively. Then match the colour produced by each sample with a colour chart. 1

(b) (i) Lemon juice 1

(ii) $10^{(6-2)} = 10^4$ 1

OR

10 000 (1)

14. (a) *x*: dilute NaOH(aq) 1

y: $Cu(OH)_2(s)$ 1

z: dilute $H_2SO_4(aq)$ 1

(b) Copper(II) ions in an aqueous solution gives a bluish green flame in the flame test. 1

15. (a) The reaction between the concentrated sulphuric acid and sodium hydroxide solution is highly exothermic. 1

OR

This is to avoid filling the burette more than once. (1)

(b) From colourless to pink 1

(c) (i) 5.0 cm^3 pipette 1

(ii) 1000.0 cm^3 volumetric flask 1

(d) Using a clean 5.0 cm^3 pipette, slowly add 5.0 cm^3 of the toilet cleaner, with constant stirring, to a beaker containing about 250 cm^3 of distilled water. 1

Then transfer the solution made, with all the washing, into a clean 1000.0 cm^3 volumetric flask. Add distilled water up to the graduation mark of the flask. Finally stopper the flask, invert it several times to mix the contents well. 1

(e) $H_2SO_4(aq) + 2NaOH(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l)$

Number of moles of NaOH used

$= 0.15 \times \frac{28.00}{1000}$ mol $= 4.2 \times 10^{-3}$ mol 1

From the equation, mole ratio of NaOH to H_2SO_4 = 2 : 1.

$\therefore$ concentration of H_2SO_4 in the 25.0 cm^3 of diluted sample of toilet cleaner

$$= \frac{\frac{4.2 \times 10^{-3}}{2}}{\frac{25.0}{1000}} \text{ mol dm}^{-3} = 0.084 \text{ mol dm}^{-3}$$ 1

Concentration of H_2SO_4 in the 5.0 cm^3 of the toilet cleaner

$= 0.084 \text{ mol dm}^{-3} \times \frac{1000.0 \text{ dm}^3}{5.0 \text{ dm}^3}$

$= 17 \text{ mol dm}^{-3}$ 1